

Persistency of the Momentum Effect

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Abstract

Momentum profits, resulting from buying winners and selling losers, are robust in the stock market worldwide. However, more than 40% of winners and losers immediately fall out of their respective groups in the month following formation, suggesting that intermediate-term momentum persistency is not universal among all stocks with extreme past performance. The return reversals that these nonpersistent winners and losers exhibit in the month following formation are strong, resulting in a monthly loss of more than 17% for a momentum strategy constructed on such stocks. By contrast, persistent winners and losers, defined as those staying in their groups for at least one more month, exhibit much stronger performance persistency. Further analysis indicates that the persistency is stronger for stocks with greater information asymmetry and more extensively heterogeneous investor beliefs, consistent with the underreaction hypothesis for price momentum.

Keywords: Delayed reaction hypothesis; Duration; Persistent losers; Persistent momentum strategy; Persistent winners; Heterogeneous beliefs; Information asymmetry

JEL Classification: G11, G14

1. Introduction

The existence of intermediate-term price momentum, as documented by Jegadeesh and Titman (1993), remains one of the most robust puzzles in financial research. Although researchers have extensively investigated the sources of the momentum effect (e.g., Lee and Swaminathan, 2000; Gutierrez and Kelley, 2008; Chen, Chen, Hsin, and Lee, 2014), what drives intermediate-term momentum persistency remains a problem to be solved.

Within a six-month formation period, approximately 47% of winner (loser) stocks fall out of their respective portfolios immediately following formation. The return reversals that such nonpersistent winners and losers exhibit in the month following formation are strong, resulting in a monthly loss of more than 17% for a momentum strategy constructed on such stocks. Overall, they exhibit weak persistency in performance. By contrast, persistent winners and losers, defined as those that stayed in their groups for at least one more month, exhibit much stronger performance persistency. This finding suggests that intermediate-term momentum persistency is not a universal phenomenon for all stocks with extreme past performance. In turn, it is of interest to explore the characteristics and driving forces that give rise to the differences in performance persistency at the individual firm level. To do so, we calculate the winner (loser) duration as the number of months that a stock consecutively stays in the winner (loser) portfolio and conduct a comprehensive duration analysis at the firm level to examine various risk-based and behavioral competing hypotheses in explaining the persistency of momentum effect.

We begin by providing a clearer understanding of the distribution of post-formation returns and the turnover rates for stocks in winner and loser portfolios. We find that at least one-quarter of winner (loser) stocks experience negative (positive) post-formation returns; thus, a momentum strategy may fail if an investor concurrently goes long for

winner stocks and shorts loser stocks. We also find that, in terms of the six-month formation period, more than 40% of winner and loser stocks are nonpersistent winners and losers. Among persistent winners (losers), 18.6% (19.0%) of those stocks have only a one-month duration, 10.8% (10.9%) have a two-month duration, and approximately 97% persist for less than six months. This large dispersion of post-formation performance and high turnover rates for winner and loser stocks inspires us to explore the difference in post-formation performance for persistent and nonpersistent winners (losers) further. To this end, we separate winner (loser) stocks into persistent and nonpersistent subsamples and examine their post-formation returns. The empirical results show that during the post-formation period, persistent winners exhibit greater price appreciation than nonpersistent winners, and persistent losers tend to lose more than nonpersistent losers.

Moreover, we demonstrate that a persistent momentum strategy, that is, buying persistent winners and selling persistent losers, can generate monthly returns as high as 1.25%, which amounts to an annual return of 15%.¹ Such a persistent momentum strategy outperforms a price momentum strategy by 19 basis points in monthly returns (an 18% improvement). The persistent momentum effect remains robust after adjusting for risk factors including the market factor, Fama-French three factors, Carhart's (1997) momentum factor, and Pastor and Stambaugh's (2003) liquidity factor. In addition, the different post-formation performance for persistent and nonpersistent momentum stocks can last for more than 6 months and does not reverse in the following 36 months, indicating that this performance difference does not result from investors' overreaction.

Among various explanations for the momentum effect,² a recent and growing body

¹ To avoid look-ahead bias and make the strategy tradable, persistent winners and losers used in the persistent momentum strategy are those defined in the month before the formation date. Section 4 provides a detailed discussion of the persistent momentum strategy.

² A variety of theories offer explanations of the momentum effect, ranging from rational accounts, such as risk-based explanations (see Chordia and Shivakumar, 2002; Sadka, 2006; Liu and Zhang, 2008) and limits to arbitrage (Arena, Haggard, and Yan, 2008), to behavioral explanations, such as overreaction (Daniel,

of literature demonstrates that the delayed reaction hypothesis can explain momentum profits. For example, Barberis, Shleifer, and Vishny (1998) develop a theoretical model suggesting that conservatism bias can lead investors to underreact to new information, and O'Hara (2003), Sadka (2006), and Chen and Zhao (2012) show that greater information asymmetry will make investors react to a firm's good or bad news more conservatively, leading to a delayed price adjustment from new information. Moreover, Allen, Morris, and Shin (2006), Hong and Stein (2007), Banerjee, Kaniel, and Kremer (2009), Verardo (2009), and Makarov and Rytchkov (2012) argue that gradual information diffusion results in greater heterogeneity among investors (hereinafter, "heterogeneous beliefs") and leads to a stronger momentum effect. The current study therefore takes advantage of the duration of persistent winners and losers to examine whether the information asymmetry and heterogeneous beliefs are associated with momentum persistency at the individual firm level. We find that, consistent with the delayed reaction hypothesis, winner and loser stocks with a higher level of information asymmetry or higher degree of heterogeneous beliefs tend to stay in their respective portfolios in the next formation period. In addition, those persistent portfolios with higher levels of information asymmetry or higher degrees of heterogeneous beliefs can generate more momentum profits in the subsequent one-year holding period.

We further conduct a duration model to investigate whether characteristics associated with information asymmetry and heterogeneous beliefs can affect winner and loser durations. We show that persistent winners and losers of relatively small size, with highly idiosyncratic volatility and low institutional ownership, will have a longer duration as winners (losers). The finding implies that investors tend to underreact to information about firms with greater information asymmetry, leading to a longer momentum effect. In

Hirshleifer, and Subrahmanyam, 1998; Cooper, Gutierrez, and Hameed, 2004) and underreaction (Antoniou, Doukas, and Subrahmanyam, 2013; Da, Gurun, and Warachka, 2014).

addition, the positive association among the duration, trading volume, and dispersion of analyst forecasts indicates that momentum stocks endure longer if investor opinions differ about their fundamental value. We also observe a strong momentum persistency for high-information asymmetry and high-heterogeneous belief stocks that exhibit a large price appreciation (depreciation). That is, the momentum effect is more persistent when stocks have more incremental information content during the formation period and those prices thus do not reflect this information efficiently. In addition, the duration model shows an asymmetric effect on momentum duration: The formation period return plays a relatively important role for persistent losers, but the post-formation return is more important for persistent winners.

Overall, this study contributes to finance literature in several ways. First, we provide evidence of the implementation cost of the momentum strategy. Knez and Ready (1996), Shleifer and Vishny (1997), Grundy and Martin (2001), and Korajczyk and Sadka (2004) argue that the trading cost may deter investors from applying the momentum strategy, though the effect of trading costs cannot fully explain momentum profits. The current study demonstrates a large dispersion of post-formation returns and a high turnover rate for winner and loser portfolios, indicating that not all winner and loser stocks follow the midterm return continuation. The momentum strategy contains an implementation risk: Investors could face a potential loss if they cannot go long for all winner stocks and go short for all loser stocks. Such an indirect implementation cost for the momentum strategy is new to the literature and presents a possible avenue to track the sources of the momentum effect.

Second, we find a stronger post-formation performance for persistent momentum portfolios and demonstrate that the persistent momentum effect cannot be explained by risk factors or the behavioral overreaction hypothesis. Prior literature demonstrates that, in terms of momentum profits, the momentum effect is associated with investors' delayed

reaction to information. However, those studies either do not mention or indirectly show that information asymmetry or heterogeneous beliefs can enhance the persistency of the momentum effect. The current study fills this research gap by introducing winner and loser duration as a measure of momentum persistency and providing a direct test of the persistency of the momentum effect at the individual firm level. We thus confirm the delayed reaction hypothesis in momentum persistency and demonstrate that information asymmetry and heterogeneous beliefs determine the persistency of the momentum effect.

Third, this study shows that a persistent momentum strategy may offer monthly returns as high as 1.25%, which outperforms Jegadeesh and Titman's (1993) price momentum strategy by 19 basis points. In addition to being of academic interest, our findings can serve as a useful guide for asset managers seeking profitable investment strategies.

The remainder of this paper is organized as follows: In Section 2, we describe the methodologies and data used herein. In Section 3, we investigate post-formation performance for persistent winners and losers. Section 4 tests the profitability of the persistent momentum strategy. In Section 5, we discuss the determinants of persistent winners and losers and conduct a duration analysis for persistent momentum portfolios. Section 6 presents the conclusions.

2. Methodology and data description

2.1 Durations for winners and losers

We first construct winner and loser portfolios according to the approach Jegadeesh and Titman (1993) suggest: At the end of each month, we rank sample stocks into deciles according to their prior J -month returns ($J = 3, 6, 9$, and 12), and group the stocks into 10 equally weighted portfolios. Sample stocks in the top decile portfolio are called

“winners,” and those in the bottom decile are called “losers.” The duration for a winner (loser) stock is defined as the number of months that a winner (loser) stock is consecutively ranked in the top (bottom) decile portfolio after the formation month. The durations for winner and stocks therefore can be expressed as

$$Duration_{i,t}^W = \sum_{k=t+1}^{t+24} I_{i,k} , \quad (1)$$

where $I_{i,k}$ is 1 if stock i is consecutively in the winner portfolio from t to k , and 0 otherwise; and

$$Duration_{i,t}^L = \sum_{k=t+1}^{t+24} I'_{i,k} , \quad (2)$$

where $I'_{i,k}$ is 1 if stock i is consecutively in the loser portfolio from t to k , and 0 otherwise.³ To avoid double-counting winner or loser duration for a firm in the same duration period, we restrict our sample firms to winner and loser stocks that do not belong to these respective portfolios in the previous formation month.

We further identify persistent winners and losers as winner and loser stocks with duration greater than 0. For example, a winner stock in month t has the highest-ranked returns from month $t - J + 1$ to month t , whereas a persistent winner in month t is defined as a stock that has the highest ranked returns for at least two consecutive periods, from $t - J + 1$ to t and from $t - J + 2$ to $t + 1$.

³ Two alternative versions of the momentum persistency based on return persistency in the post-formation period yielded similar results. First, we applied return persistency, defined as the number of positive (negative) monthly returns for a winner (loser) stock out of 24 post-formation months. Second, we applied consecutive return persistency, defined as the number of consecutive positive (negative) monthly returns for a winner (loser) after the formation month. In unreported results, both alternative measures provided similar results in the following analyses.

2.2 Information asymmetry and heterogeneous beliefs

This study examines whether factors associated with the delayed reaction hypothesis can determine the durations of winner and loser stocks. Atiase (1985) and Huddart and Ke (2007) suggest that firms with smaller capitalization or lower institutional ownership may suffer more serious information asymmetry problems, which leads investors to react to the new information about such a firm more conservatively. Therefore, firms with a higher degree of information asymmetry exhibit a stronger momentum return (e.g., Zhang, 2006; Chen and Zhao, 2012). In contrast, Harris and Raviv (1993), Shalen (1993), Gebhardt, Lee, and Swaminathan (2001), Boehme, Danielsen, and Sorescu (2006), Hong and Stein (2007), and Verardo (2009) show that idiosyncratic volatility and/or trading volume can represent the degree of investors' heterogeneous beliefs. Their findings indicate that if a firm has a higher trading volume or higher idiosyncratic volatility, the large divergence of investors' opinions about the firm's intrinsic value will lead to a stronger return continuation of the firm in the midterm. However, those studies only investigate the relative strength of the momentum effect in terms of the magnitude of post-formation returns rather than the persistency of the momentum effect. The duration analysis offers an opportunity to examine whether factors associated with information asymmetry and heterogeneous beliefs can determine the persistency of the momentum effect—more specifically, the duration for persistent winners and losers.

To test the effect of information asymmetry on momentum persistency, we use idiosyncratic volatility (e.g., Roll, 1988; Morck, Yeung, and Yu, 2000; Ferreira and Laux, 2007) and institutional ownership (e.g., Shiller and Pound, 1989; Huddart and Ke, 2007) as proxies for the level of information asymmetry. We define idiosyncratic volatility as the residual variance from regressing a firm's daily excess returns on market daily excess returns over the past year. Institutional ownership is the percentage of outstanding shares

held by institutional investors. To test the effect of heterogeneous beliefs, we include monthly stock turnover (e.g., Hong and Stein, 2007), standardized unexplained volume (e.g., Garfinkel and Sokobin, 2006), change in turnover (e.g., Garfinkel and Sokobin, 2006), and dispersion of analyst earnings forecasts (e.g., Diether, Malloy and Scherbina, 2002) as proxies for investors' disagreement. We define a firm's monthly turnover as the average daily ratio of the number of shares traded to the number of shares outstanding.⁴ The standardized unexplained volume (*SUV*) is a standardized prediction error from a regression of trading volume on the absolute value of stock returns.⁵ The change in turnover (*DTO*) is the difference between a firm's average daily market-adjusted turnover for a certain month and the average daily market-adjusted turnover over the previous year, where the market-adjusted turnover ratio is the difference of a firm's turnover ratio and the market turnover ratio in the same month. We define the dispersion of analyst earnings forecasts as the standard deviation of analysts' earnings forecasts for a firm.

2.3 Data

We collect basic firm information and firm accounting data from COMPUSTAT and obtain stock price, stock returns, trading volume, market capitalization, share codes, and exchange codes from the Center for Research in Security Prices (CRSP). Institutional holdings are compiled quarterly from SEC 13-F filings. Quarterly analyst forecasts are from the I/B/E/S Detail History files. Because I/B/E/S data consist of sufficient quarterly analyst earnings forecasts beginning after 1980, we select the sample period 1980–2013. The sample includes only common stocks (*SHRCD* = 10, 11) and firms traded on the

⁴ Gould and Kleidon (1994) show that trading volume for NASDAQ securities may be inflated due to the double counting of dealer trades. We therefore follow Anderson and Dyl (2005) and scale down the trading volume for NASDAQ securities by 50% before 1997 and by 38% after 1997.

⁵ We use one-month daily prediction errors from the regression of daily trading volume on the absolute values of stock returns over the past year. The regression can be expressed as

$Turnover_{i,t} = a_i + b_{i,1} |R_{i,t}^+| + b_{i,1} |R_{i,t}^-| + e_{i,t}$. The standardized unexplained volume is the ratio of the average of daily prediction errors to the standard deviation of the daily prediction errors.

NYSE, AMEX, or NASDAQ. We exclude closed-end funds, real estate investment trusts (REITs), American depository receipts (ADRs), utility services, and financial institutions. We also exclude firms with stock prices below \$5 on the formation date, as investors generally pay only limited attention to such stocks. To obtain durations for winner and loser stocks and their price performance, we limit our sample to firms that had at least six consecutive monthly returns prior to the formation month. In addition, data for firm-months must include their corresponding size, book-to-market ratio, idiosyncratic volatility, institutional ownership, monthly stock turnover, standardized unexplained volume, change in turnover, and dispersion of analyst earnings forecasts in each formation month.

To ensure that firm accounting information was available to the public investors at the time the stock returns were recorded, we follow Fama and French's (1992) approach and match the accounting data for the fiscal year ending in calendar year $t - 1$ with the returns for July of year t to June of $t + 1$. We calculate the market capitalization by multiplying the closing price on the last trading day of June of that year by the number of outstanding shares at the end of June of that year. We calculate the book-to-market ratio as a firm's book value of common equity for the fiscal year ending in year $t - 1$ divided by its market equity at the end of December in year $t - 1$. To avoid potential distortion by outliers, we winsorize the variables at 2.5% and 97.5%, including market capitalization, book-to-market rate, idiosyncratic volatility, institutional ownership, monthly stock turnover, standardized unexplained volume, change in turnover, dispersion of analyst earnings forecasts, prior price return, and post-formation return.⁶

Table 1 shows the descriptive statistics of momentum profits and major variables for winner and loser portfolios. In terms of six-month post-formation performance for

⁶ The main results for the analyses are similar whether we winsorize or trimmed those variables at 2.5% and 97.5% or 5% and 95%.

winner and loser stocks based on their prior six-month returns ($J = 6$), winner stocks exhibit an average 0.49% price appreciation, with a standard deviation as high as 3.19%, and loser stocks suffer a 0.96% loss, with a standard deviation of 4.15%. In addition, the lower quartile performance for winner stocks is -1.33% , and the upper quartile performance for loser stocks is 1.38% . Consistent with Lesmond, Schill, and Zhou (2004), the finding of large variation in post-formation performance for winners and losers indicates that not all winner and loser stocks exhibit midterm return continuity. On the contrary, at least one-quarter of winner and loser stocks appear to have a contrarian pattern in the initial months following the formation month. Given the limited number of stocks an investor or a fund manager can trade (see Sapp and Yan, 2008), buying a portion of winner stocks and selling a portion of loser stocks may not yield a positive return and may even result in a potential loss.⁷ Therefore, though a price momentum strategy can statistically generate a significantly positive return, high variations of the post-formation performance for winner and loser stocks may lead to a potential loss if investors randomly long and short for a small portion of winner and loser stocks.

[Insert Table 1 about here]

Table 2 presents distributions of the duration for winner and loser stocks in different formation periods. For the six-month formation period, 46.69% of winners and 47.00% of losers have zero duration, and the remaining 53.31% of winners and 53.00% of losers are persistent winners and persistent losers. Approximately 19% of winners and losers can only stay in momentum portfolios for two consecutive months (duration is equal to one month), approximately 11% of winners and losers can stay in momentum portfolios for three consecutive months (duration is equal to two months), and

⁷ Sapp and Yan (2008) show that equity mutual funds hold only 90.85 stocks on average, and those in the top quintile hold an average of 228.83 stocks. Therefore, it is impossible for an investor or a portfolio manager to buy all winner stocks and sell all loser stocks to obtain the average momentum profits, as shown in prior literature.

approximately 95% of winners and losers cannot stay in momentum portfolios longer than seven months (duration is equal to six months). The average durations are 0.889 months for all winners and 1.914 months for persistent winners, while the average durations are 0.836 months for all losers and 1.855 months for persistent losers. The finding of short durations for winner and loser stocks supports Table 1's finding of high variations of post-formation performance for momentum stocks.

[Insert Table 2 about here]

3. Persistent and nonpersistent momentum portfolios

3.1 Turnover rates of winner and loser portfolios

Table 3 summarizes the average turnover rates for winner and loser portfolios for different formation periods. For the six-month formation period, 52.83% of winner stocks and 52.96% of loser stocks will stay in their respective groups in the following month, and 1.42% of winner stocks become losers and 1.58% of loser stocks become winners just one month later.⁸ We consider a turnover to occur when a winner (loser) stock is not ranked as the top (bottom) decile if we retire its oldest monthly return and include its most recent monthly return one month after the formation date. The observation that a change of only one-sixth of the formation period return will knock nearly 48% of winner and loser stocks out of their respective groups confirms the importance of the role of the subsequent monthly returns. This finding is consistent with our observation in Table 1 and indicates the existence of large variation in post-formation performance between persistent winners (losers) and nonpersistent winners (losers). The large variation and

⁸ One may argue that nonpersistent winner stocks are those with lower formation period returns, such that they can easily fall from the winner portfolio the next month. We therefore further divide sample firms into 20 groups in each formation month and investigate turnover rates for extreme winner and loser portfolios. Appendix Table A.1 shows that only 32.45% of top 5% winners and 49.13% of bottom 5% losers stay in extreme portfolios, indicating that the persistency of momentum portfolios is not mainly driven by extreme previous formation period performance.

high turnovers therefore inspire us to further explore price behaviors for persistent and nonpersistent momentum portfolios.

[Insert Table 3 about here]

3.2 Post-formation performance of persistent and nonpersistent momentum portfolios

We divide winner (loser) stocks into a persistent winner (loser) group and a nonpersistent winner (loser) group and examine their post-formation performance. Panel A of Table 4 presents the averages of 13 subsequent monthly returns after the formation month t for persistent and nonpersistent winners. The winner stocks exhibit an average monthly return (R_{t+1}) of 2.67% one month after the formation month; however, a closer examination reveals that persistent winners have a monthly return as high as 7.34% and nonpersistent winners have a loss of 4.87%.⁹ In addition, in terms of raw return and risk-adjusted return, we observe significantly different post-formation performance between persistent and nonpersistent winners up to six months after the formation month. It indicates that the better performance for persistent winners is not due to the higher monthly returns one month after the formation date for persistent winners or the severe short-term reversal effect for nonpersistent winners.

Similarly, Panel B of Table 3 shows that in loser portfolios, relative to nonpersistent losers, persistent losers experience large losses one month after the formation month, and this relatively bad price performance can last for eight months after the formation month. In addition, the unreported analysis shows that the strong (weak) post-formation performance for persistent winners (losers) relative to nonpersistent winners (losers) can be observed in different formation periods ($J = 3$, $J = 9$, and $J = 12$).

⁹ Our results still hold when we skip one week in calculating post-formation returns to avoid the bid-ask bounce.

[Insert Table 4 about here]

Table 4 also provides some interesting insights regarding the implementation of a price momentum strategy. As Jegadeesh and Titman (1993) suggest, price momentum strategies usually skip a week between the formation period and the holding period, to avoid the short-term reversal effect induced by a bid-ask bounce, price pressure, and the lagged reaction effects documented in Jegadeesh (1990) and Lehmann (1990). Stocks with strong short-term reversal are typically treated as nonpersistent winners or losers. However, Table 4 indicates that, even though skipping one week in a price momentum strategy can avoid a potential loss from the short-term reversal, investors or fund managers who include nonpersistent winners or losers in their momentum portfolios still may obtain relatively weak momentum profits in the subsequent holding period.

4. Persistent momentum effect

4.1 Profitability of persistent momentum strategies

The previous section shows a large spread of post-formation returns between persistent and nonpersistent momentum portfolios, suggesting a higher momentum profit from the persistent momentum strategy by taking a long position for persistent winners, taking a short position for persistent losers, and holding the portfolio for the subsequent six months.¹⁰ Specifically, at the end of month t , we form a zero investment portfolio with a long position for stocks with the top decile return from the end of $t - 6$ to the end of t and a short position for stocks with the bottom decile return from the end of $t - 6$ to the end of t . We hold this zero investment portfolio for the next six months with a one-month skip between formation period and holding period, and do not rebalance it during the holding period. Panel A of Table 5 provides raw and adjusted returns for the

¹⁰ The persistent momentum strategy uses information about the stock return from $t - 6$ to t and is free of look-ahead bias.

persistent momentum strategy, the nonpersistent momentum strategy, and the price momentum strategies suggested by Jegadeesh and Titman (1993). In terms of the six-month formation period and the six-month holding period, a persistent momentum strategy yields a monthly raw return as high as 1.25%, while the returns earned by a nonpersistent momentum strategy and traditional six-month and seven-month formation period momentum strategies are 0.47%, 1.06%, and 1.09%, respectively. Note that profits from nonpersistent momentum strategy as well as those from price momentum strategies can be explained by Fama-French three factors and Carhart's momentum factor, whereas persistent momentum strategies can sustain significantly positive profits after adjusting for those factors.

[Insert Table 5 about here]

Table 5, Panel B, presents return differences between the persistent momentum strategy and various momentum strategies. We find that investors can earn an average monthly return of 1.25% by buying persistent winner stocks and selling persistent loser stocks, a strategy that outperforms the price momentum strategy by 19 basis points. The superior performance for a persistent momentum strategy still holds after adjusting for various risk factors. The price momentum strategy incorporates prior six-month returns to define winners and losers, while the persistent momentum strategy requires the prior seven months' returns to identify persistent winners and losers. To counter the argument that the persistent momentum strategy's superior performance is a result of its inclusion of one more month of information than the price momentum strategy, we also conduct a price momentum strategy with a seven-month formation period and obtain a monthly raw return of 1.09%. Compared with the persistent momentum strategy's yields of 1.25%, the seven-month formation period price momentum strategy still cannot perform as well.

We also apply the cross-sectional regressions George and Hwang (2004) suggest to simultaneously isolate the returns contributed by persistent and nonpersistent momentum portfolios:

$$R_{i,t} = a_{j,t} + b_{1j,t} R_{i,t-1} + b_{2j,t} \ln(\text{size})_{i,t-1} + b_{3j,t} DW_{i,t-j-1} + b_{4j,t} DL_{i,t-j-1} + b_{5j,t} DPW_{i,t-j} + b_{6j,t} DPL_{i,t-j} + \varepsilon_{i,t}, \quad (3)$$

where $j = 1, \dots, 6$. We regress firm i 's return in month t on four portfolio dummies. We include prior one-month return and the logarithm of market capitalization in the regression to control for bid-ask bounce effect and size effect. The momentum portfolio dummy, $DW_{i,t-j-1}$ ($DL_{i,t-j-1}$), equals 1 if firm i is ranked as a winner (loser) in month $t - j - 1$, and 0 otherwise. The persistent momentum portfolio dummy, $DPW_{i,t-j}$ ($DPL_{i,t-j}$), indicates whether firm i is included in the persistent winner (loser) portfolio in month $t - j$. Estimated coefficients of portfolio dummies are averaged over $j = 1, \dots, 6$. Table 6 presents time-series averages of the six-month averages for each momentum portfolio. In terms of raw return, average coefficients for $DW - DL$ and $DPW - DPL$ are 0.11% and 1.62%, with significance levels of 5% and 1%, respectively. Specifically, the average coefficient for $DW - DL$ is the monthly return contributed by the information content of momentum portfolios, which is unrelated to the information content of the persistent momentum portfolio. Therefore, the average coefficient of 0.11% for $DW - DL$ is the monthly return to a nonpersistent momentum strategy. Similar to Table 5, the nonpersistent momentum strategy can only generate a few profits with marginal significance. In contrast, the average coefficient for $DPW - DPL$, which is the isolated return contributed by the persistent momentum portfolios, is higher. This higher isolated return indicates that the persistent momentum strategy can generate more profits than the nonpersistent momentum strategy. In addition, we find that the isolated return for the persistent momentum strategy in non-January months is higher than that in all months,

while the return for the nonpersistent momentum strategy in non-January months has no significant difference from that in all months. The January seasonality for the momentum strategy, as found by Jegadeesh and Titman (1993), Odean (1998), and Grinblatt and Moskowitz (2004), thus appears confined to persistent momentum portfolios.

In summary, the lower return for the nonpersistent momentum strategy and higher isolated return for the persistent momentum strategy can be observed for not only raw returns but also returns adjusted for various risk factors. Tables 5 and 6 confirm the significant profitability of the persistent momentum strategy and demonstrate that the persistent momentum effect is not driven by price momentum or higher risk loadings.

[Insert Table 6 about here]

4.2 Can overreaction explain the persistent momentum effect?

Recent behavioral models (e.g., Barberis, Shleifer, and Vishny, 1998; Daniel, Hirshleifer, and Subrahmanyam, 1998; Hong and Stein, 1999) propose the overreaction hypothesis to explain the momentum effect, suggesting that the representative heuristic, investor's self-attribution, and expectations conditional on partial information may result in a short-term momentum and a long-term price correction. If the profitability of persistent momentum effect can also be explained by the overreaction hypothesis, a long-term reversal for returns from persistent momentum strategy will be observed. We therefore investigate the isolated return contributed by the persistent momentum portfolios from the cross-sectional regressions for the first-, second-, and third-year holding periods. As Table 7 shows, the isolated return to the persistent momentum strategy is significantly positive for the first-year holding period, indicating that the persistent momentum strategy is profitable and outperforms nonpersistent momentum strategy by an average monthly return of 0.89% including January and 1.02% excluding January. The isolated returns to the persistent momentum strategy for the first-year

holding period are significant after controlling for Fama-French three factors, but the superior performance for the persistent momentum portfolio disappears after the first year. Except a negative raw return with a marginal significance observed in the second-year holding period, no significant reversal occurs for the isolated return to the persistent momentum strategy in the second and third years. The positive return for the persistent momentum effect thus can be observed in the first-year post-formation period and will not reverse for up to three years. Inconsistent with the traditional momentum effect, the profitability of the persistent momentum strategy will not reverse in the long run, and therefore, the persistent momentum effect cannot be explained by the overreaction hypothesis.

[Insert Table 7 about here]

4.3 Can a delayed price adjustment model explain the persistent momentum effect?

An alternative behavioral model posits that investors' conservatism in the face of asymmetric information and heterogeneous beliefs about a stock's fundamental value may lead to a longer price adjustment period. If such a delayed reaction hypothesis can explain the profitability of the persistent momentum strategy, a higher persistent momentum return will be observed for persistent winners with a higher level of information asymmetry or more sharply heterogeneous beliefs among investors. We therefore conduct the cross-sectional regressions with two additional interaction terms, as indicated in the following.

$$R_{i,t} = a_{j,t} + b_{1j,t}R_{i,t-1} + b_{2j,t}\ln(\text{size})_{i,t-1} + b_{3j,t}DW_{i,t-j-1} + b_{4j,t}DL_{i,t-j-1} + b_{5j,t}DPW_{i,t-j} + b_{6j,t}DPL_{i,t-j} + b_{7j,t}DPW_{i,t-j} \times D_{\text{delayed_reaction}_{i,t-j}} + b_{8j,t}DPL_{i,t-j} \times D_{\text{delayed_reaction}_{i,t-j}} + \varepsilon_{i,t}, \quad (4)$$

where $j = 1, \dots, 6$, and $D_{\text{delayed_reaction}_{i,t-j}}$ is a dummy variable equal to 1 if stock i 's information asymmetry (or heterogeneous beliefs) measure is in the top decile in month t

– *j*. We use idiosyncratic volatility (*Ivol*) and institutional ownership (*IO*) to proxy for information asymmetry, and we use monthly turnover (*Turnover*), standardized unexplained volume (*SUV*), change in turnover (*DTO*), and dispersion in analysts' forecasts (*Dispersion*) to proxy for heterogeneous beliefs. If persistent momentum profits are generated by the delayed price adjustment due to information asymmetry (or heterogeneous beliefs) for persistent winners and losers, a higher coefficient for the interaction term between the dummy of persistent winner and the dummy of high information asymmetry (or of high heterogeneous beliefs) and a lower coefficient for the interaction term between the dummy of persistent loser and that of high information asymmetry (or of high heterogeneous beliefs) will be observed. As Panel A of Table 8 shows, persistent winners with higher information asymmetry can generate higher profits than persistent winners that do not have an information asymmetry problem, while persistent losers with higher information asymmetry exhibit greater losses than other persistent losers. We also find that persistent winners (losers) with more heterogeneous beliefs will experience higher (lower) future returns in the following 12 months. Panel B shows similar results when we exclude January returns. We conclude that, consistent with the delayed reaction hypotheses, the persistent momentum effect is mainly driven by persistent momentum stocks with information asymmetry or heterogeneous beliefs.

[Insert Table 8 about here]

5. Duration analysis for persistent momentum portfolios

5.1 Determinants of persistent momentum stocks

In light of the previous finding that better post-formation performance for persistent momentum stocks is associated with information asymmetry and heterogeneous beliefs, we subsequently investigate whether the delayed reaction hypothesis can determine the

persistence of momentum effect. Specifically, we use a cross-sectional probit regression at the individual firm level to examine which factors can turn a winner (loser) stock into a persistent winner (loser). The model specification is denoted as

$$P\left(\text{Duration}_{i,t}^W > 0\right) = a_t + b_{1,t} \ln(\text{Size})_{i,t} + b_{2,t} \text{BM}_{i,t} + b_{3,t} \text{Ivol}_{i,t} + b_{4,t} \text{IO}_{i,t} + b_{5,t} \text{Turnover}_{i,t} + b_{6,t} \text{SUV}_{i,t} + b_{7,t} \text{DTO}_{i,t} + b_{8,t} \text{Dispersion}_{i,t} + b_{9,t} R_{i,t-6,t} + b_{10,t} R_{i,t,t+1} + \varepsilon_{i,t} \quad \text{and} \quad (5)$$

$$P\left(\text{Duration}_{i,t}^L > 0\right) = a_t + b_{1,t} \ln(\text{Size})_{i,t} + b_{2,t} \text{BM}_{i,t} + b_{3,t} \text{Ivol}_{i,t} + b_{4,t} \text{IO}_{i,t} + b_{5,t} \text{Turnover}_{i,t} + b_{6,t} \text{SUV}_{i,t} + b_{7,t} \text{DTO}_{i,t} + b_{8,t} \text{Dispersion}_{i,t} + b_{9,t} R_{i,t-6,t} + b_{10,t} R_{i,t,t+1} + \varepsilon_{i,t}, \quad (6)$$

where $\text{Duration}_{i,t}^W$ and $\text{Duration}_{i,t}^L$ are the durations for winner and loser stocks, respectively, and the dependent variable is a dummy indicating whether a winner (loser) stock is persistent; that is, the dependent variable is 1 if a winner (loser) stock has a nonzero duration, and 0 otherwise. The independent variables are the natural logarithm of the market capitalization, book-to-market ratio, idiosyncratic volatility, institutional ownership, monthly turnover, standardized unexplained volume, change in turnover, previous six-month return, and one-month post-formation return.

Table 9 presents the time-series averages of estimates from the probit regressions. The average coefficients on idiosyncratic volatility and institutional ownership are significantly positive and negative for both winner and loser stocks, indicating that both winner and loser firms with higher idiosyncratic volatility and lower institutional ownership are more likely to be persistent winners and losers. That is, investors will underreact to new information for winner and loser firms with higher levels of information asymmetry, leading to a more persistent momentum effect. However, for the average coefficients on market capitalization, which can also proxy for information asymmetry, we only find significant results in support of the role of information asymmetry in winner groups. One possible explanation is that we have excluded firms with a stock price below \$5 from our sample; low-priced loser stocks may exhibit a

certain period of price depreciation and be classified as nonpersistent losers. Hence, we may have been unable to observe a significantly negative relationship between momentum persistency and market capitalization in the loser group.

The significantly positive slopes on monthly stock turnover, standardized unexplained trading volume, change in volume, and dispersion of analyst forecasts confirm the role of heterogeneous beliefs in explaining the momentum persistency. That is, for winner and loser firms, firms with a higher degree of disagreement on firm values among investors or analysts are more likely to stay in their respective groups in the next formation period. Table 9 also provides evidence that both the formation-period return ($R_{t-6, t}$) and the one-month return subsequent to the formation month ($R_{t, t+1}$) are important in determining persistent winners and losers. This finding is consistent with Table 4, in which the significance of post-formation performance implies that being a persistent winner or loser stock does not depend solely on its previous higher or lower returns but also on its price performance after the formation period.

[Insert Table 9 about here]

5.2 Determinants of the duration for persistent momentum stocks

In this section, we focus on persistent winners and losers to examine whether information asymmetry or heterogeneous beliefs can determine the duration of persistent momentum portfolios. Specifically, we investigate whether variables associated with information asymmetry or investor disagreement can affect the duration of momentum stocks. We also examine the role of formation and post-formation performance in determining the duration of persistent momentum stocks. We apply a duration model with censored data suggested by Ongena and Smith (2001) and Deville and Riva (2007)

to address the dependent variable, the duration of persistent winners and losers, censored between 0 and 24.¹¹ The model specification is denoted as

$$\begin{aligned} Duration_{i,t} = & a_t + b_{1,t} \ln(Size)_{i,t} + b_{2,t} BM_{i,t} + b_{3,t} Ivol_{i,t} + b_{4,t} IO_{i,t} + b_{5,t} Turnover_{i,t} + b_{6,t} SUV_{i,t} \\ & + b_{7,t} DTO_{i,t} + b_{8,t} Dispersion_{i,t} + b_{9,t} |R_{i,t-6,t}| + b_{10,t} |R_{i,t-6,t}| \times Dw_{i,t} \\ & + b_{11,t} |R_{i,t,t+1}| + b_{12,t} |R_{i,t,t+1}| \times Dw_{i,t} + \varepsilon_{i,t} \end{aligned} \quad (7)$$

where the dependent variable is the duration for persistent momentum stocks, and the independent variables are the natural logarithm of market capitalization, book-to-market ratio, idiosyncratic volatility, institutional ownership, monthly turnover, standardized unexplained volume, change in turnover, dispersion of analyst forecasts, absolute value of previous six-month return, absolute value of one-month post-formation return, interaction term between absolute value of previous six-month return and winner dummy, and interaction term between absolute value of one-month post-formation return and winner dummy. The two interaction terms allow us to investigate whether performance during the formation period or post-formation period has a different effect on the momentum duration between persistent winners and persistent losers. Table 10 presents the time-series averages of estimates from the duration models. They show that momentum duration is positively associated with idiosyncratic volatility and negatively associated with institutional ownership. If persistent winner and loser stocks have relatively higher idiosyncratic volatilities and lower institutional ownership, those stock prices will underreact to new information due to investors' conservatism, and the slow price adjustment to new information causes those persistent winner and loser stocks to remain in their respective groups longer. Moreover, the finding that momentum duration

¹¹ Estimations of the ordinary least square would be biased if the dependent variable violates the normality assumption. Because the dependent variable in Eq. (7), the duration of momentum portfolios, is between 0 and 24, we adopt a duration model to investigate the influence of explanatory variables on the nonnegative and right-censored dependent variable, assuming the duration follows a Weibull distribution. We determine parameter estimates and standard errors using the standard optimization algorithm. We also perform the estimation with the specification of exponential distribution and obtain similar results.

is positively associated with turnover variables and dispersion of analyst forecasts supports the role of heterogeneous beliefs in explaining the duration of persistent momentum portfolios: Heterogeneous beliefs also cause winner and loser stock prices to inefficiently reflect new information, thus resulting in longer duration of persistent winners and losers. Consistent with the delayed reaction hypothesis, the results in Table 10 show that higher levels of information asymmetry and heterogeneous beliefs will slow the speed of price adjustment for persistent momentum stocks, thus resulting in longer durations for persistent momentum stocks.

[Insert Table 10 about here]

Table 10 also shows that both formation period performance and one-month post-formation returns can determine the duration of persistent winners and losers. The significantly positive coefficients indicate that the higher formation period returns and higher monthly return after the formation date can extend the duration of persistent winner and loser stocks. Moreover, we find a slightly negative coefficient on the interaction between the winners dummy and formation period performance and a significantly positive coefficient on the interaction between the winners dummy and one-month post-formation return. The asymmetry effect on the duration between persistent winners and losers indicates that, in terms of the duration of momentum portfolios, post-formation performance plays a more important role for persistent winners than persistent losers, and formation period performance is relatively less important for persistent winners than for persistent losers.

5.3 Delayed price adjustment for persistent momentum portfolios

As discussed in the previous section, we find that formation period performance is positively associated with the persistency of momentum effect. One possible explanation for this finding is the delayed price adjustment to new information. If firm-specific

information goes public (e.g., revenue announcement, earnings announcement, security offerings), investors will evaluate the firm's value on the basis of that new information and give it a new price. Therefore, a large price appreciation for a firm during the formation period may indicate that the firm had good news during the recent period, and vice versa. However, if the market is not efficient enough to fully reflect the incremental information contents conveyed by the good or bad news, a longer duration following a large price appreciation or depreciation during the formation period can be observed. Another possible explanation for the longer duration following a large price movement is the definition of the winner (loser) duration: When we calculate formation period returns to assign stocks as winners (losers) at each formation date, we retire their oldest monthly returns from the formation returns one month before the formation date and include their most recent monthly return. This calculation points to a positive relationship between formation period performance and duration for those firm-months with a duration of six or less (if $J = 6$). Therefore, to examine whether the duration of persistent momentum portfolios can be determined by the delayed price adjustment, we add interaction terms of prior six-month returns with a variable associated with information asymmetry or heterogeneous beliefs. The model specification is denoted as

$$\begin{aligned}
 Duration_{i,t} = & a_t + b_{1,t} \ln(Size)_{i,t} + b_{2,t} BM_{i,t} + b_{3,t} Ivol_{i,t} + b_{4,t} IO_{i,t} + b_{5,t} Turnover_{i,t} \\
 & + b_{6,t} |R_{i,t,t+1}| + b_{7,t} |R_{i,t-6,t}| + b_{8,t} |R_{i,t-6,t}| \times Turnover_{i,t} \\
 & + b_{9,t} |R_{i,t-6,t}| \times Ivol_{i,t} + b_{10,t} |R_{i,t-6,t}| \times IO_{i,t} + \varepsilon_{i,t} .
 \end{aligned} \tag{8}$$

By introducing the interaction terms, we effectively separate the effect of delayed price adjustment and the definition of duration for momentum stocks. The coefficients of interaction terms of prior six-month return with a variable associated with information asymmetry or heterogeneous beliefs can represent the effect of the slow price adjustment due to information asymmetry or heterogeneous beliefs among investors.

Table 11 presents the time-series averages of estimates from the duration model based on Eq. (8). Our variables of interest are prior six-month return and the interaction terms. Significantly positive coefficients for the prior six-month return indicate that the duration of persistent momentum portfolios will be longer if persistent momentum portfolios exhibit a higher return during the formation period. Such a positive relationship mostly results from the definition of duration for momentum stocks, in that a large monthly return during the formation period can affect the duration up to six months after the formation date. Table 11 also provides evidence of delayed price adjustment for persistent winners and losers. Significant coefficients of the interaction terms indicate that if the persistent momentum portfolios have more incremental information content during the formation period and suffer greater information asymmetry or heterogeneous beliefs, the stock prices of persistent momentum portfolios may need more time to fully reflect the incremental information contents, and therefore a more persistent momentum effect can be observed.¹²

[Insert Table 11 about here]

6. Conclusions

We conduct a comprehensive analysis of the persistence of the momentum effect. Only 60% of winner and loser stocks can be persistent winners and losers, and at least 25% of winner and loser stocks experience a contrarian rather than a momentum effect. Such high turnover rates and large return dispersions in the post-formation period for momentum stocks indicate that the momentum strategy not only involves a higher direct

¹² To rule out the direct effect of the definition of duration, we also examine the information diffusion for persistent momentum portfolios with duration greater than J , which is the length of the formation period. However, due to the limited number of observations, we use fixed-effects censored regressions instead of cross-sectional duration models. As presented in Appendix Table A.2, significant interaction terms indicate that slow information diffusion holds though the reduction of observations, and the introduction of time-series effect weakens the effect of slow information diffusion.

transaction cost but also contains an indirect implementation cost, in that momentum investors may face a loss if they cannot include all winner and loser stocks in their portfolio.

We find that in the post-formation period, persistent winner stocks outperform nonpersistent winner stocks, while persistent loser stocks continuously underperform compared with nonpersistent loser stocks. The finding of a large spread of post-formation returns between persistent and nonpersistent momentum portfolios confirms the importance of persistence for momentum portfolios in determining the strength of the momentum effect. In addition, this pattern indicates a zero-investment strategy of buying persistent winners and selling persistent losers. We propose an enhanced trading strategy, the persistent momentum strategy, which yields an average monthly return of 1.25% and remains significant after adjusting for various asset pricing factors.

The delayed reaction hypothesis provides a behavioral explanation for the momentum effect, showing that investors' conservatism in the face of asymmetric information and heterogeneous beliefs about a firm's liquidation value may lead to a delayed reaction to the stock price of the company. The results of the duration analysis provide new evidence that both information asymmetry and heterogeneous beliefs can explain not only the magnitude of momentum profits but also the persistency of the momentum effect. We also find an asymmetric effect, in that the post-formation return contributes to the winner persistency more, and the formation period return can better explain loser persistency.

As mentioned previously, though the momentum effect has been robust and persistent in the stock market for more than four decades, the persistency of winner and loser stocks is less well understood. The current study contributes to extant finance literature by presenting new evidence of market efficiency with respect to momentum

persistence. However, further research is necessary to trace the distribution and persistence of post-formation performance for momentum portfolios.

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Table 1. Summary statistics for momentum returns and major variables

This table presents summary statistics for the returns of momentum strategies and major variables of the sample. Momentum return is the return from momentum strategy based on the previous J -month return ($J = 3, 6, 9$, and 12) and 6-month holding period. Size is a stock's market capitalization. Book-to-market ratio (BM) is defined as a firm's book common equity for the fiscal year ending in year $t - 1$ divided by its market equity at the end of December in year $t - 1$. Turnover is the monthly stock turnover ratio of the monthly trading volume to the total number of shares outstanding. The standardized unexplained volume (SUV) is a standardized prediction error calculated by a regression of trading volume on the absolute value of stock returns. The change in turnover (DTO) is the difference between a firm's average daily market-adjusted turnover for the month and the average daily market-adjusted turnover over the previous year, where the market-adjusted turnover ratio is the difference of a firm's turnover ratio and the market turnover ratio in the same month. The table also reports the past J -month returns ($R_{t-J+1,t}$) and 1-month return subsequent to the formation month ($R_{t,t+1}$). The idiosyncratic volatility ($Ivol$) is the residual variance from regressing a firm's daily excess returns on market daily excess returns over the previous 12 months. Institutional ownership (IO) is the percentage of outstanding shares held by institutional investors. Dispersion is defined as the standard deviation in analysts' forecasts for stocks in each month. Panel A presents the winner portfolios, and Panel B presents the loser portfolios.

Panel A. Winner portfolios

	Mean	Std.	Min	Q1	Median	Q3	Max
$J = 3$ Momentum return (%)	0.3507	3.2301	-12.0795	-1.4821	0.5906	2.2311	10.4870
Size (\$mil.)	667.4588	1764.6061	7.2358	37.2350	125.7143	446.9844	12421.1360
BM	0.8284	0.6678	0.0924	0.3366	0.6273	1.1068	2.7209
Turnover	1.3971	1.4231	0.0322	0.3235	0.8306	2.0029	4.8831
SUV	0.4514	0.9425	-1.9185	-0.1937	0.5118	1.1680	1.9215
DTO	0.9471	0.4640	-0.0450	0.7547	0.9703	1.1622	2.0020
$R_{t-J+1,t}$ (%)	12.1258	4.4272	-3.8643	8.8384	12.3911	17.1919	17.1920
$R_{t,t+1}$ (%)	1.8135	13.1531	-21.6973	-6.8694	0.5427	9.1736	31.8462
Ivol	0.3570	1.5482	0.0073	0.0234	0.0312	0.0421	10.0128
IO (%)	0.3735	0.2861	<0.001	0.1173	0.3245	0.5971	0.9343
Dispersion	0.1284	0.2191	0.0040	0.0177	0.0480	0.1312	1.1818
$J = 6$ Momentum return (%)	0.4874	3.1934	-12.2785	-1.3306	0.7227	2.3531	9.2244
Size (\$mil.)	642.9134	1722.7996	7.2358	37.3119	122.6694	424.5774	12421.1360
BM	0.8527	0.6839	0.0924	0.3445	0.6463	1.1460	2.7209
Turnover	1.5834	1.5090	0.0322	0.3916	1.0029	2.3663	4.8831
SUV	0.6021	0.9209	-1.9185	-0.0132	0.6775	1.3218	1.9215
DTO	0.9068	0.4984	-0.0450	0.6793	0.9404	1.1472	2.0020
$R_{t-J+1,t}$ (%)	9.2835	2.8522	-1.1054	7.2103	9.8893	12.0638	12.0638
$R_{t,t+1}$ (%)	2.0567	13.6913	-21.6971	-7.1059	0.7600	10.0002	31.8462
Ivol	0.3864	1.6118	0.0073	0.0251	0.0328	0.0438	10.0128
IO (%)	0.3654	0.2857	<0.001	0.1095	0.3121	0.5848	0.9343
Dispersion	0.1238	0.2126	0.0040	0.0158	0.0452	0.1258	1.1818
$J = 9$ Momentum return (%)	0.4588	3.2816	-13.0366	-1.5186	0.7020	2.4255	10.3599
Size (\$mil.)	669.8437	1757.7363	7.2358	39.4800	131.5280	452.0995	12421.1360
BM	0.8437	0.6849	0.0924	0.3344	0.6321	1.1363	2.7209
Turnover	1.6612	1.5405	0.0322	0.4215	1.0762	2.5290	4.8831
SUV	0.6433	0.9117	-1.9185	0.0369	0.7232	1.3651	1.9215
DTO	0.9000	0.5156	-0.0450	0.6504	0.9368	1.1517	2.0020
$R_{t-J+1,t}$ (%)	7.7588	2.2186	-0.3171	6.1732	8.2386	9.8987	9.8987
$R_{t,t+1}$ (%)	2.1167	13.8500	-21.6973	-7.2623	0.7892	10.3453	31.8462
Ivol	0.3750	1.5782	0.0073	0.0253	0.0329	0.0439	10.0128
IO (%)	0.3735	0.2867	<0.001	0.1158	0.3237	0.5991	0.9343
Dispersion	0.1233	0.2139	0.0040	0.0152	0.0438	0.1244	1.1818
$J = 12$ Momentum ret (%)	0.3610	3.2875	-13.0861	-1.5841	0.6280	2.4166	10.5813
Size (\$mil.)	706.4927	1829.2011	7.2358	41.3488	139.4400	480.7585	12421.1360
BM	0.8249	0.6771	0.0924	0.3252	0.6154	1.1093	2.7209
Turnover	1.6894	1.5515	0.0322	0.4318	1.1041	2.5878	4.8831
SUV	0.6552	0.9034	-1.9185	0.0527	0.7339	1.3688	1.9215
DTO	0.9096	0.5280	-0.0450	0.6500	0.9432	1.1675	2.0020
$R_{t-J+1,t}$ (%)	6.8253	1.8480	-0.6386	5.5203	7.2346	8.5990	8.5990
$R_{t,t+1}$ (%)	2.0634	13.8691	-21.6965	-7.3801	0.8055	10.4174	31.8462
Ivol	0.3668	1.5562	0.0073	0.0253	0.0328	0.0435	10.0128
IO (%)	0.3786	0.2875	<0.001	0.1188	0.3311	0.6083	0.9343
Dispersion	0.1194	0.2083	0.0040	0.0141	0.0424	0.1202	1.1818

Panel B. Losers portfolios

		Mean	Std.	Min	Q1	Median	Q3	Max
<i>J</i> = 3	Momentum return (%)	−0.8682	3.8872	−16.7264	−2.8204	−0.5686	1.4565	10.8404
	Size (\$mil.)	554.8794	1574.7406	7.2356	34.2240	107.7420	364.1160	12421.0243
	BM	0.6856	0.5870	0.0923	0.2749	0.5052	0.8900	2.7209
	Turnover	1.2582	1.3552	0.0322	0.2811	0.7072	1.7276	4.8834
	SUV	0.2268	0.9162	−1.9185	−0.4148	0.2584	0.8979	1.9215
	DTO	1.0343	0.4575	−0.0450	0.8430	1.0378	1.2434	2.0020
	$R_{t-J+1,t}$ (%)	−8.4432	3.6563	−13.0481	−12.3168	−8.4393	−5.5054	1.4244
	$R_{t,t+1}$ (%)	2.7426	13.7692	−21.6974	−6.3911	1.4488	10.8701	31.8663
	Ivol	0.3144	1.4494	0.0073	0.0232	0.0313	0.0422	10.0128
	IO (%)	0.3834	0.2782	0.0000	0.1411	0.3401	0.5960	0.9343
	Dispersion	0.1614	0.2533	0.0040	0.0255	0.0658	0.1681	1.1818
<i>J</i> = 6	Momentum return (%)	−0.9572	4.1511	−15.8818	−2.9308	−0.7334	1.3836	19.1072
	Size (\$mil.)	497.5325	1444.6139	7.2356	34.4131	104.3594	331.9876	12421.0243
	BM	0.6243	0.5575	0.0923	0.2453	0.4506	0.7989	2.7209
	Turnover	1.3892	1.4156	0.0322	0.3310	0.8244	1.9611	4.8834
	SUV	0.2792	0.9094	−1.9185	−0.3490	0.3204	0.9447	1.9215
	DTO	1.0206	0.4898	−0.0450	0.8123	1.0292	1.2526	2.0020
	$R_{t-J+1,t}$ (%)	−6.4105	2.5043	−9.0358	−9.0360	−6.7921	−4.4306	1.8566
	$R_{t,t+1}$ (%)	3.0929	14.5187	−21.6974	−6.8965	1.7205	12.2443	31.8663
	Ivol	0.3139	1.4457	0.0073	0.0254	0.0336	0.0448	10.0128
	IO (%)	0.3833	0.2751	0.0000	0.1469	0.3398	0.5891	0.9343
	Dispersion	0.1794	0.2725	0.0040	0.0285	0.0734	0.1874	1.1818
<i>J</i> = 9	Momentum return (%)	−0.9030	4.1133	−14.8928	−2.9063	−0.6601	1.3904	19.4745
	Size (\$mil.)	471.7119	1380.6491	7.2356	34.2669	101.0255	312.8604	12421.0243
	BM	0.6068	0.5472	0.0923	0.2378	0.4368	0.7724	2.7209
	Turnover	1.3947	1.4144	0.0322	0.3392	0.8304	1.9662	4.8834
	SUV	0.2627	0.9064	−1.9185	−0.3643	0.2974	0.9231	1.9215
	DTO	1.0210	0.4946	−0.0450	0.8092	1.0300	1.2609	2.0020
	$R_{t-J+1,t}$ (%)	−5.1269	2.0183	−7.2233	−7.2233	−5.4402	−3.4872	1.5310
	$R_{t,t+1}$ (%)	3.1405	14.6659	−21.6974	−6.9765	1.6948	12.5000	31.8663
	Ivol	0.3092	1.4289	0.0073	0.0260	0.0343	0.0456	10.0128
	IO (%)	0.3811	0.2731	0.0000	0.1486	0.3356	0.5823	0.9343
	Dispersion	0.1884	0.2815	0.0040	0.0300	0.0766	0.1996	1.1818
<i>J</i> = 12	Momentum return (%)	−0.7352	4.0852	−16.1833	−2.4661	−0.5597	1.3685	21.6841
	Size (\$mil.)	464.4828	1365.9930	7.2356	34.7710	102.0240	309.1258	12421.0243
	BM	0.6159	0.5503	0.0923	0.2430	0.4459	0.7850	2.7209
	Turnover	1.3847	1.4055	0.0322	0.3381	0.8256	1.9456	4.8834
	SUV	0.2404	0.9071	−1.9185	−0.3882	0.2712	0.8963	1.9215
	DTO	1.0220	0.4908	−0.0450	0.8120	1.0299	1.2599	2.0020
	$R_{t-J+1,t}$ (%)	−4.3213	1.6937	−6.0927	−6.0927	−4.5941	−2.929	1.5230
	$R_{t,t+1}$ (%)	3.1669	14.7122	−21.6969	−7.0183	1.6876	12.5000	31.8663
	Ivol	0.3014	1.4078	0.0073	0.0262	0.0343	0.04553	10.0128
	IO (%)	0.3842	0.2731	<0.001	0.1531	0.3401	0.5849	0.9343
	Dispersion	0.1973	0.2900	0.0040	0.0321	0.0804	0.2098	1.1818

Table 2. Distribution of winner/loser durations

This table presents the percentage and the number of observations for winner and loser durations. Duration is the length of time that a winner (loser) stock consecutively stays in the winner (loser) portfolio in the subsequent month(s). To avoid the overlapping problem, observations included are firm-months that were not in a winner or loser portfolio in the previous month. Panel A shows duration distributions for winner portfolios, and Panel B shows duration distributions for loser portfolios.

Panel A: Winner portfolios

Duration	<i>J</i> = 3		<i>J</i> = 6		<i>J</i> = 9		<i>J</i> = 12	
	Obs.	%	Obs.	%	Obs.	%	Obs.	%
0	32,406	52.87	20,739	46.69	17,685	46.59	17,054	48.60
1	12,798	20.88	8,248	18.57	6,315	16.64	5,125	14.61
2	10,750	17.54	4,811	10.83	3,693	9.73	3,287	9.37
3	2,841	4.64	3,070	6.91	2,408	6.34	2,016	5.75
4	1,289	2.10	2,103	4.74	1,791	4.72	1,476	4.21
5	620	1.01	2,526	5.69	1,255	3.31	1,072	3.06
6	303	0.49	1,086	2.45	986	2.60	886	2.53
7	139	0.23	633	1.43	776	2.04	678	1.93
8	73	0.12	407	0.92	1,086	2.86	582	1.66
9	35	0.06	271	0.61	596	1.57	478	1.36
10	24	0.04	171	0.39	389	1.03	432	1.23
11	8	0.01	116	0.26	265	0.70	640	1.82
12	2	<0.01	88	0.20	180	0.47	381	1.09
13	2	<0.01	46	0.10	136	0.36	241	0.69
14	0	0.00	32	0.07	110	0.29	178	0.51
15	0	0.00	25	0.06	68	0.18	117	0.33
16	0	0.00	15	0.03	52	0.14	95	0.27
17	1	<0.01	11	0.03	53	0.14	58	0.17
18	0	0.00	7	0.02	30	0.08	79	0.23
19	0	0.00	3	0.01	25	0.07	48	0.14
20	0	0.00	6	0.01	13	0.03	31	0.09
21	0	0.00	1	<0.01	19	0.05	25	0.07
22	0	0.00	0	0.00	9	0.02	21	0.06
23	0	0.00	2	0.01	7	0.02	24	0.07
24	0	0.00	0	0.00	15	0.04	65	0.19
<i>Including Duration=0</i>	<i>J</i> = 3		<i>J</i> = 6		<i>J</i> = 9		<i>J</i> = 12	
Mean	0.8890		1.5139		1.9098		2.1521	
Std.	1.2604		2.2701		3.0792		3.4015	
<i>Excluding Duration=0</i>	<i>J</i> = 3		<i>J</i> = 6		<i>J</i> = 9		<i>J</i> = 12	
Mean	1.9143		3.0163		3.8394		4.4949	
Std.	1.2013		2.3984		3.4295		4.2726	

Panel B: Loser portfolios

	<i>J</i> = 3		<i>J</i> = 6		<i>J</i> = 9		<i>J</i> = 12	
Duration	Obs.	%	Obs.	%	Obs.	%	Obs.	%
0	33,543	53.69	21,648	47.00	18,452	46.15	17,949	48.11
1	13,069	20.92	8,748	18.99	6,784	16.97	5,639	15.12
2	10,569	16.92	5,021	10.90	4,136	10.34	3,562	9.55
3	3,007	4.81	3,137	6.81	2,631	6.58	2,254	6.04
4	1,300	2.08	2,178	4.73	1,855	4.64	1,555	4.17
5	600	0.96	2,767	6.01	1,383	3.46	1,204	3.23
6	217	0.35	1,124	2.44	1,045	2.61	998	2.68
7	82	0.13	548	1.19	842	2.11	795	2.13
8	52	0.08	371	0.81	1,271	3.18	622	1.67
9	19	0.03	199	0.43	573	1.43	536	1.44
10	8	0.01	128	0.28	336	0.84	433	1.16
11	3	0.01	76	0.17	237	0.59	703	1.88
12	2	<0.01	54	0.12	130	0.33	325	0.87
13	1	<0.01	24	0.05	94	0.24	222	0.60
14	0	0.000	15	0.03	77	0.19	162	0.43
15	0	0.000	11	0.02	46	0.12	87	0.23
16	0	0.000	4	0.01	35	0.09	61	0.16
17	0	0.000	2	<0.01	19	0.05	52	0.14
18	0	0.000	1	<0.01	19	0.05	59	0.16
19	0	0.000	0	0.00	9	0.02	28	0.08
20	0	0.000	0	0.00	7	0.02	18	0.05
21	0	0.000	0	0.00	2	0.01	14	0.04
22	0	0.000	1	<0.01	1	<0.01	9	0.02
23	0	0.000	0	0.00	1	<0.01	4	0.01
24	0	0.000	0	0.00	0	0.00	16	0.04
<i>Including Duration=0</i>	<i>J</i> = 3		<i>J</i> = 6		<i>J</i> = 9		<i>J</i> = 12	
Mean	0.8364		1.4264		1.8052		2.0533	
Std.	1.2001		2.1008		2.8202		3.4015	
<i>Excluding Duration=0</i>	<i>J</i> = 3		<i>J</i> = 6		<i>J</i> = 9		<i>J</i> = 12	
Mean	1.8554		2.8342		3.5649		4.1671	
Std.	1.1106		2.1519		3.0447		3.8173	

Table 3. Turnover rates for winner and loser stocks

This table reports the average turnover rates for winner and loser stocks in the next formation month. At the end of each month, winner stocks are those stocks with top decile previous J -month ($J = 3, 6, 9$, and 12) returns, while loser stocks are those stocks with bottom decile previous J -month returns. Panel A presents average percentages of winner stocks that will be ranked and assigned to a certain decile in the next formation month. Panel B presents average percentages of loser stocks that will be ranked and assigned to a certain decile in the next formation month.

Panel A: Winner portfolios

J	P1 (Losers)	P2	P3	P4	P5	P6	P7	P8	P9	P10 (Winners)
3	1.52%	1.66%	1.89%	2.28%	2.86%	4.11%	6.04%	10.31%	20.91%	46.64%
6	1.42%	1.25%	1.27%	1.48%	1.93%	2.81%	4.54%	8.85%	23.67%	52.83%
9	1.62%	1.37%	1.33%	1.40%	1.87%	2.39%	3.81%	8.03%	25.31%	52.96%
12	1.96%	1.56%	1.48%	1.71%	2.01%	2.62%	3.93%	7.52%	25.88%	49.54%

Panel B: Loser portfolios

J	P1 (Losers)	P2	P3	P4	P5	P6	P7	P8	P9	P10 (Winners)
3	46.11%	20.74%	10.30%	6.14%	4.25%	3.11%	2.54%	2.10%	1.84%	1.69%
6	52.96%	24.06%	9.03%	4.31%	2.82%	1.87%	1.57%	1.40%	1.24%	1.58%
9	52.69%	25.44%	7.82%	3.83%	2.44%	1.75%	1.46%	1.20%	1.39%	1.75%
12	50.36%	25.97%	7.66%	3.57%	2.50%	1.82%	1.57%	1.60%	1.64%	2.05%

Table 4. Average post-formation returns for persistent and non-persistent momentum stocks

This table reports average monthly post-formation returns for persistent and nonpersistent momentum stocks. At the end of each month, winner stocks are those stocks with top decile previous six-month returns, while loser stocks are those stocks with bottom decile previous six-month returns. Persistent winners (losers) are those winner (loser) that stocks stay in the winner (loser) group both in the formation month t and in the next formation month $t+1$, while nonpersistent winners (losers) are those winner (loser) stocks that do not stay in the winner (loser) group in the next formation month $t + 1$. Average monthly post-formation returns and risk-adjusted returns by Fama-French three-factor model (FF-3) are reported. Panel A presents the subsequent 13 monthly equally weighted returns for winners, persistent winners, and nonpersistent winners. Panel B presents the subsequent 13 monthly equally weighted returns for losers, persistent losers, and nonpersistent losers. All values are presented as percentages. *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively.

Panel A: Winner portfolios

Portfolio	R_{t+1}	R_{t+2}	R_{t+3}	R_{t+4}	R_{t+5}	R_{t+6}	R_{t+7}	R_{t+8}	R_{t+9}	R_{t+10}	R_{t+11}	R_{t+12}	R_{t+13}
Winner	2.6713	2.5250	1.4840	1.3916	1.3818	1.3568	1.3044	1.0588	0.9163	0.9163	0.8542	0.6584	0.5003
Persistent winner	7.3353	2.5642	1.7125	1.5159	1.4980	1.4884	1.3719	1.1907	0.9502	0.8838	0.8355	0.7015	0.5187
Nonpersistent winner	-4.8683	2.5054	1.1180	1.1691	1.1565	1.1454	1.1987	0.8200	0.8740	0.9608	0.8606	0.6186	0.4527
Persistent – Nonpersistent	12.2036*** (33.20)	0.0589 (1.21)	0.5945*** (4.07)	0.3468*** (2.85)	0.3415** (2.54)	0.3431*** (2.92)	0.1733 (1.23)	0.3707*** (2.79)	0.0762 (0.58)	-0.0770 (-0.54)	-0.0251 (-0.20)	0.0829 (0.59)	0.0659 (0.56)
FF-3 adjusted	11.9988*** (34.47)	0.1436 (1.46)	0.3335*** (3.84)	0.3037** (2.51)	0.3438** (2.31)	0.1680** (2.22)	-0.0061 (0.71)	0.2890** (2.41)	0.1833 (0.55)	0.0369 (-0.42)	-0.1840 (0.01)	-0.0213 (0.78)	-0.0037 (0.45)

Panel B. Loser portfolios

Portfolio	R_{t+1}	R_{t+2}	R_{t+3}	R_{t+4}	R_{t+5}	R_{t+6}	R_{t+7}	R_{t+8}	R_{t+9}	R_{t+10}	R_{t+11}	R_{t+12}	R_{t+13}
Loser	4.7382	-0.3615	-0.2857	-0.1058	0.0329	0.0605	-0.0371	0.0382	0.2594	0.3081	0.5405	0.7135	1.0472
Persistent loser	-2.3319	-0.8914	-0.4405	-0.3932	-0.2368	-0.1832	-0.2014	-0.2221	-0.1832	0.0709	0.2643	0.5303	0.9414
Nonpersistent loser	12.4872	0.0999	-0.1125	0.2101	0.3237	0.2567	0.1510	0.3224	0.6830	0.5922	0.9090	0.8958	1.1496
Persistent – Nonpersistent	-14.8190*** (-30.71)	-0.9913*** (3.86)	-0.3280 (-1.56)	-0.6033*** (-2.85)	-0.5606** (-2.49)	-0.4399* (-1.77)	-0.3524* (-1.68)	-0.5445** (-2.55)	-0.8662*** (-4.20)	-0.5213** (-2.46)	-0.6447*** (-2.96)	-0.3655 (-1.62)	-0.2082 (-0.95)
FF-3 adjusted	-14.8341*** (-30.10)	-0.1034*** (-5.41)	-0.2923 (-1.42)	-0.0554 (-0.25)	-0.366 (-1.24)	-0.4944** (-2.42)	-0.5081** (-2.15)	-0.4332** (-2.34)	-0.2601 (-1.42)	-0.0517 (-0.20)	0.1828 (0.77)	0.5163** (2.19)	0.4447** (2.07)

Table 5. Returns to persistent momentum strategies

This table presents returns in percentages and the associated *t*-statistics from persistent momentum strategies, nonpersistent momentum strategies, and Jegadeesh and Titman's (1993) momentum strategies executed during the period from 1980 to 2013. At the end of each month, winner stocks are those stocks with top decile previous six-month returns, while loser stocks are those stocks with bottom decile previous six-month returns. Persistent winners (losers) are those winner (loser) stocks that stay in the winner (loser) group both in formation month $t - 1$ and in the formation month t , while nonpersistent winners (losers) are those winner (loser) stocks in formation month $t - 1$ that do not stay in the winner (loser) group in formation month t . For the persistent momentum strategy, the zero-investment portfolios that long for persistent winner stocks and short for persistent loser stocks are held for six subsequent months with one-month skip from the formation date and are not rebalanced during the holding period. For the nonpersistent momentum strategy, portfolios of buying nonpersistent winner stocks and selling nonpersistent loser stocks are held for six subsequent months with one-month skip from the formation date and not rebalanced during the holding period. For the Jegadeesh and Titman's (1993) momentum strategy, portfolios are developed by buying winner stocks and selling loser stocks and are held for six subsequent months with one-month skip from the formation date and not rebalanced during the holding period. Panel A shows the average monthly returns for persistent, nonpersistent, and Jegadeesh and Titman's (1993) momentum strategies based on the six-month formation period return, and Jegadeesh and Titman's (1993) momentum strategies based on the seven-month formation period return ($J = 7$). Panel B reports the average monthly return differences between persistent momentum strategies and other types of momentum strategies. Raw returns and risk-adjusted returns for various momentum strategies and the differences between momentum strategies are presented in this table. Risk-adjusted returns are intercepts of regressions of the capital asset price model (CAPM), the Fama-French three-factor model (FF-3), three Fama-French factors plus Carhart's momentum factor (FF-4), three Fama-French factors plus Pastor and Stambaugh's liquidity factor (FF-3+LIQ), and five Fama-French factors (FF-5). The associated Newey-West *t*-statistics are in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively.

Panel A. Returns to persistent, non-persistent, and JT momentum strategies

Persistent Momentum	Raw	CAPM	FF-3	FF-4	FF-3+LIQ	FF-5
Loser	0.5792 (1.17)	-0.2910 (-0.89)	-0.2900 (-0.98)	0.4302** (2.00)	-0.2808 (-0.95)	0.2439 (0.84)
Winner	1.8315*** (4.78)	1.1005*** (4.98)	1.1630*** (6.12)	0.7535*** (4.96)	1.1700*** (6.15)	1.2297*** (9.00)
Winner – Loser	1.2552*** (3.35)	1.3915*** (3.74)	1.4530*** (3.85)	0.3233* (1.65)	1.4508*** (3.84)	0.9858** (2.58)
Nonpersistent Momentum	Raw	CAPM	FF-3	FF-4	FF-3+LIQ	FF-5
Loser	0.9362** (1.96)	0.1424 (0.42)	0.1003 (0.33)	0.6707** (2.56)	0.1135 (0.37)	0.6838** (2.35)
Winner	1.4068*** (4.09)	0.7629*** (3.71)	0.7196*** (4.07)	0.4363*** (2.73)	0.7248*** (4.09)	0.8282*** (7.33)
Winner – Loser	0.4706 (1.48)	0.6205** (1.97)	0.6193** (1.96)	-0.2344 (-1.15)	0.6113* (1.93)	0.1444 (0.45)
JT Mom. ($J = 6$)	Raw	CAPM	FF-3	FF-4	FF-3+LIQ	FF-5
Loser	0.6953 (1.46)	-0.1638 (-0.54)	-0.1601 (-0.60)	0.5047*** (2.61)	-0.1491 (-0.55)	0.3327 (1.26)
Winner	1.7554*** (4.79)	1.0511*** (5.04)	1.0890*** (6.06)	0.7073*** (4.88)	1.0952*** (6.08)	1.1549*** (9.48)
Winner – Loser	1.0588*** (3.08)	1.2149*** (3.57)	1.2491*** (3.63)	0.2026 (1.18)	1.2443*** (3.60)	0.8221** (2.35)
JT Mom. ($J = 7$)	Raw	CAPM	FF-3	FF-4	FF-3+LIQ	FF-5
Loser	0.6906 (1.44)	-0.173 (-0.56)	-0.1759 (-0.64)	0.5145*** (2.62)	-0.1635 (-0.59)	0.3199 (1.18)
Winner	1.7845*** (4.86)	1.078*** (5.17)	1.1278*** (6.28)	0.7327*** (5.17)	1.1349*** (6.31)	1.1842*** (9.44)
Winner – Loser	1.0940*** (3.11)	1.2499*** (3.59)	1.3037*** (3.70)	0.2182 (1.30)	1.2984*** (3.68)	0.8642** (2.41)

Panel B. Returns differences between persistent momentum strategies and other types of momentum strategies

Persistent Momentum vs. JT Momentum ($J = 6$)	Raw	CAPM	FF-3	FF-4	FF-3+LIQ	FF-5
Loser	-0.1162** (-2.45)	-0.1272*** (-2.68)	-0.1299*** (-2.70)	-0.0745 (-1.61)	-0.1317*** (-2.73)	-0.0888* (-1.78)
Winner	0.0773*** (2.70)	0.0494* (1.89)	0.0740*** (2.98)	0.0464* (1.94)	0.0748*** (3.01)	0.0749*** (2.81)
Winner – Loser	0.1934*** (3.28)	0.1766*** (2.99)	0.2039*** (3.43)	0.1208** (2.18)	0.2065*** (3.47)	0.1637*** (2.63)
Persistent Momentum vs. JT Momentum ($J = 7$)	Raw	CAPM	FF-3	FF-4	FF-3+LIQ	FF-5
Loser	-0.1114** (-2.43)	-0.1180** (-2.55)	-0.1141** (-2.45)	-0.0843* (-1.80)	-0.1173** (-2.50)	-0.0760 (-1.57)
Winner	0.0469* (1.84)	0.0223 (0.97)	0.0352 (1.54)	0.0208 (0.91)	0.0351 (1.55)	0.0456* (1.94)
Winner – Loser	0.1583*** (2.94)	0.1403*** (2.61)	0.1493*** (2.74)	0.1052* (1.94)	0.1524*** (2.79)	0.1216** (2.14)
Persistent Mom. vs. Nonpersistent Mom.	Raw	CAPM	FF-3	FF-4	FF-3+LIQ	FF-5
Loser	-0.3570** (2.04)	-0.4320** (-2.49)	-0.3903** (-2.22)	-0.2405 (-1.39)	-0.3943** (-2.24)	-0.4398** (-2.44)
Winner	0.4247*** (4.19)	0.3386*** (3.55)	0.4433*** (4.85)	0.3172*** (3.72)	0.4452*** (4.86)	0.4015*** (4.21)
Winner – Loser	0.7817*** (4.21)	0.7706*** (4.12)	0.8337*** (4.42)	0.5577*** (3.20)	0.8395*** (4.44)	0.8413*** (4.30)

Table 6. Comparison of persistent momentum and non-persistent momentum strategies

This table presents returns to relative strength portfolios and momentum strategies. Each month during the period from 1980 through 2013, six cross-sectional regressions are estimated for momentum, and persistent momentum strategies:

$$R_{i,t} = a_{j,t} + b_{1j,t} R_{i,t-1} + b_{2j,t} \ln(\text{size})_{i,t-1} + b_{3j,t} D_{W_{i,t-j}} + b_{4j,t} D_{L_{i,t-j}} + b_{5j,t} D_{PW_{i,t-j-1}} + b_{6j,t} D_{PL_{i,t-j-1}} + \varepsilon_{i,t},$$

where $R_{i,t}$ and $\text{size}_{i,t}$ are the monthly return and the market capitalization of stock i in month t ; and $D_{W_{i,t-j-1}}$ ($D_{L_{i,t-j-1}}$) is the winner (loser) dummy that takes the value of 1 if stock i is ranked as winner (loser) in month $t-j-1$, and zero otherwise. $D_{PW_{i,t-j}}$ ($D_{PL_{i,t-j}}$) is the persistent winner (loser) dummy that takes the value of 1 if stock i is ranked as winner (loser) in both month $t-j$ and month $t-j-1$, and 0 otherwise. The estimated coefficients of independent variable are averaged over $j = 1, \dots, 6$. Average coefficients for all months and January excluded are presented. The numbers reported for raw returns are the time-series average of these averages. The t -statistics calculated from the time series are in parentheses. Risk-adjusted returns are intercepts of regressions of the Fama-French three-factor model (FF-3). The associated Newey-West t -statistics are in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively.

	Raw		FF-3	
	Jan. Inc.	Jan. Exc.	Jan. Inc.	Jan. Exc.
<i>Intercept</i>	1.7312*** (3.57)	0.7925 (1.49)	0.9918*** (3.76)	0.1527 (0.55)
R_{t-1}	-5.2714*** (-11.51)	-3.1584*** (-7.55)	-4.9298*** (-12.41)	-3.0245*** (-7.57)
$\ln(\text{Size})$	-0.1344*** (-2.60)	0.0151 (0.31)	-0.1095*** (-2.68)	0.0091 (0.23)
D_{PW}	0.6011*** (4.03)	0.6680*** (3.53)	0.7026*** (5.66)	0.7224*** (4.84)
D_{PL}	-0.9045*** (-6.54)	-0.9805*** (-5.42)	-1.0038*** (-7.75)	-1.1223*** (-7.46)
D_W	0.0492* (1.77)	0.0617* (1.75)	0.0559* (1.76)	0.0529 (1.49)
D_L	-0.0617* (-1.80)	-0.0778* (-1.77)	-0.0369 (-1.01)	-0.0648 (-1.50)
$D_{PW} D_{PL}$	1.6165*** (6.92)	1.7881*** (6.09)	1.7992*** (7.68)	1.9625*** (7.29)
$D_W - D_L$	0.1109** (2.50)	0.1395** (2.33)	0.0928* (1.90)	0.1177** (2.05)

Table 7. Long-term profits for persistent momentum strategies

This table presents long-term holding period returns to relative strength portfolios and momentum strategies. Each month during the period from 1980 through 2013, 12 cross-sectional regressions are estimated for momentum, and persistent momentum strategies:

$$R_{i,t} = a_{j,t} + b_{1j,t}R_{i,t-1} + b_{2j,t}\ln(\text{size})_{i,t-1} + b_{3j,t}D_{W_{i,t-j}} + b_{4j,t}D_{L_{i,t-j}} + b_{5j,t}D_{PW_{i,t-j-1}} + b_{6j,t}D_{PL_{i,t-j-1}} + \varepsilon_{i,t},$$

where $R_{i,t}$ and $\text{size}_{i,t}$ are the monthly return and the market capitalization of stock i in month t ; and $D_{W_{i,t-j}}$ ($D_{L_{i,t-j}}$) is the winner (loser) dummy that takes the value of 1 if stock i is ranked as winner (loser) in month $t-j-1$, and zero otherwise. $D_{PW_{i,t-j}}$ ($D_{PL_{i,t-j}}$) is the persistent winner (loser) dummy that takes the value of 1 if stock i is ranked as winner (loser) in both month $t-j$ and month $t-j-1$, and 0 otherwise. The estimated coefficients of independent variable are averaged over $j = 1, \dots, 12$, $j = 13, \dots, 24$, and $j = 25, \dots, 36$. Average coefficients for all months and January excluded are presented. The numbers reported for raw returns are the time-series average of these averages. The t -statistics calculated from the time series are in parentheses. Risk-adjusted returns are intercepts of regressions of the Fama-French three-factor model (FF-3). The associated Newey-West t -statistics are in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively.

	Year1				Year2				Year3			
	Raw		FF-3		Raw		FF-3		Raw		FF-3	
	Jan. Inc.	Jan. Exc.	Jan. Inc.	Jan. Exc.	Jan. Inc.	Jan. Exc.	Jan. Inc.	Jan. Exc.	Jan. Inc.	Jan. Exc.	Jan. Inc.	Jan. Exc.
<i>Intercept</i>	0.9143*	0.0772	0.2135	-0.5580**	1.0202**	0.9081*	0.313	0.2679	1.1390**	0.9779*	0.4227	0.3324
	(1.76)	(0.15)	(0.73)	(-2.13)	(1.99)	(1.73)	(1.10)	(0.99)	(2.24)	(1.90)	(1.52)	(1.25)
R_{t-1}	-3.0549***	-2.4375***	-2.8864***	-2.2855***	-3.3122	-3.4665***	-3.1191***	-3.3071***	-3.478***	-3.5691***	-3.2577***	-3.3979***
	(-7.55)	(-5.97)	(-6.90)	(-5.73)	(-8.03)	(-8.37)	(-7.36)	(-8.23)	(-8.39)	(-8.67)	(-7.76)	(-8.50)
$\ln(\text{Size})$	0.0370	0.1512***	0.0385	0.1452***	0.0247	0.0097	0.0252	0.0013	0.0091	0.0047	0.0096	-0.0042
	(0.76)	(3.24)	(0.94)	(3.93)	(0.52)	(0.20)	(0.63)	(0.03)	(0.19)	(0.10)	(0.25)	(-0.11)
D_{PW}	0.2900*	0.3159**	0.3269***	0.3592***	-0.1562	-0.1802	-0.1164	-0.1615*	-0.0576	-0.0372	-0.0806	-0.0778
	(1.92)	(1.98)	(2.76)	(3.06)	(-1.46)	(-1.51)	(-1.40)	(-1.88)	(-0.47)	(-0.29)	(-0.85)	(-0.79)
D_{PL}	-0.5872***	-0.6028***	-0.7396***	-0.7548***	0.1372	-0.0493	-0.0335	-0.2051*	0.1923	-0.0273	0.0738	-0.1335
	(-3.86)	(-3.90)	(-5.55)	(-5.78)	(0.98)	(-0.35)	(-0.31)	(-1.80)	(1.50)	(-0.20)	(0.82)	(-1.47)
D_W	0.0552**	0.0500*	0.0466*	0.0410	-0.0072	-0.0087	-0.0161	-0.0227	-0.0067	-0.0232	-0.0058	-0.0194
	(1.98)	(1.84)	(1.68)	(1.49)	(-0.23)	(-0.25)	(-0.53)	(-0.71)	(-0.26)	(-0.74)	(-0.20)	(-0.59)
D_L	-0.0401	-0.0360	-0.0351	-0.0322	0.0113	-0.0198	0.0075	-0.0165	-0.0188	0.0064	-0.0173	0.0077
	(-1.43)	(-1.24)	(-1.19)	(-1.10)	(0.35)	(-0.59)	(0.24)	(-0.49)	(-0.59)	(0.20)	(-0.55)	(0.25)
$D_{PW} - D_{PL}$	0.9725***	1.0047***	1.1481***	1.1871***	-0.3119*	-0.1198	-0.1064	0.0374	-0.2379	-0.0396	-0.1429	0.0287
	(4.07)	(4.29)	(5.17)	(5.43)	(-1.72)	(-0.67)	(-0.65)	(0.23)	(-1.42)	(-0.23)	(-1.03)	(0.21)
$D_W - D_L$	0.0953**	0.0860**	0.0816**	0.0732*	-0.0185	0.0111	-0.0236	-0.0062	0.0121	-0.0296	0.0114	-0.0270
	(2.50)	(2.22)	(2.03)	(1.81)	(-0.45)	(0.24)	(-0.60)	(-0.15)	(0.32)	(-0.72)	(0.27)	(-0.62)

Table 8. Delayed price adjustment for persistent momentum strategies

This table presents returns to relative strength portfolios. Each month during the period from 1980 through 2013, six cross-sectional regressions are estimated for momentum and persistent momentum strategies:

$$R_{i,t} = a_{j,t} + b_{1,j,t} R_{i,t-1} + b_{2,j,t} \ln(\text{size})_{i,t-1} + b_{3,j,t} DW_{i,t-j-1} + b_{4,j,t} DL_{i,t-j-1} + b_{5,j,t} DPW_{i,t-j} + b_{6,j,t} DPL_{i,t-j} \\ + b_{7,j,t} DPW_{i,t-j} \times D_{\text{delayed_reaction}_{i,t-j}} + b_{8,j,t} DPL_{i,t-j} \times D_{\text{delayed_reaction}_{i,t-j}} + \varepsilon_{i,t}$$

where $R_{i,t}$ and $\text{size}_{i,t}$ are the monthly return and the market capitalization of stock i in month t ; and $DW_{i,t-j-1}$ ($DL_{i,t-j-1}$) is the winner (loser) dummy that takes the value of 1 if stock i is ranked as winner (loser) in month $t-j-1$, and 0 otherwise. $DPW_{i,t-j}$ ($DPL_{i,t-j}$) is the persistent winner (loser) dummy that takes the value of 1 if stock i is ranked as winner (loser) in both month $t-j$ and month $t-j-1$, and 0 otherwise. $D_{\text{delayed_reaction}_{i,t-j}}$ is a dummy variable equal to 1 if stock i 's information asymmetry (or heterogeneous beliefs) measure is among top decile in month $t-j$. Information asymmetry measures include idiosyncratic volatility ($Ivol$) and institutional ownership (IO), and heterogeneous measures includes monthly turnover ($Turnover$), standardized unexplained volume (SUV), change in turnover (DTO), and dispersion in analysts' forecasts ($Dispersion$). The estimated coefficients of independent variable are averaged over $j = 1, \dots, 6$. The numbers reported for raw returns are the time-series average of these averages. The t -statistics calculated from the time series are in parentheses. Risk-adjusted returns are intercepts of regressions of the Fama-French 3-factor model (FF-3). The associated Newey-West t -statistics are in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively. Panel A presents average coefficients of cross-sectional regressions including all months. Panel B presents average coefficients of cross-sectional regressions excluding January.

Panel A. The interaction terms between delayed-reaction factors and dummy of persistent winner/loser: January included

	Raw	FF-3	Raw	FF-3	Raw	FF-3	Raw	FF-3	Raw	FF-3	Raw	FF-3
D_{PW}	0.4202*** (4.13)	0.4907*** (4.88)	0.6317*** (3.43)	0.6681*** (4.49)	0.4826*** (4.23)	0.5722*** (5.13)	0.8438*** (4.61)	0.8769*** (5.82)	0.5094*** (2.85)	0.5479*** (3.71)	0.5212*** (2.83)	0.5619*** (3.96)
D_{PL}	-0.8946*** (-6.40)	-0.9988*** (-7.62)	-0.9106*** (-5.40)	-1.0569*** (-6.59)	-1.0445*** (-8.61)	-1.1430*** (-9.99)	-0.7943*** (-4.90)	-0.9369*** (-6.13)	-1.1123*** (-6.48)	-1.2557*** (-7.61)	-0.8549*** (-5.38)	-0.9937*** (-6.67)
$D_{PW} \times D_{Ivol}$	0.2344** (2.17)	0.2780*** (3.21)										
$D_{PL} \times D_{Ivol}$	-0.6533*** (-3.61)	-0.6450*** (-3.71)										
$D_{PW} \times D_{IO}$			-0.4626 (-1.63)	-0.4373* (-1.71)								
$D_{PL} \times D_{IO}$			-0.7999** (-1.99)	-0.5817 (-1.56)								
$D_{PW} \times D_{TR}$					0.0451 (0.27)	0.0933 (0.65)						
$D_{PL} \times D_{TR}$					-0.3879** (-2.28)	-0.3893** (-2.55)						
$D_{PW} \times D_{SUV}$							0.3427*** (4.46)	0.3596*** (5.24)				
$D_{PL} \times D_{SUV}$							-0.7604*** (-5.88)	-0.7747*** (-5.80)				
$D_{PW} \times D_{DTO}$									0.3091** (2.17)	0.2861** (2.26)		
$D_{PL} \times D_{DTO}$									-0.1622 (-1.58)	-0.1418 (-1.46)		
$D_{PW} \times D_{Dispersion}$											0.5542*** (3.14)	0.4153** (2.56)
$D_{PL} \times D_{Dispersion}$											-0.6553*** (-3.03)	-0.8258*** (-4.20)

Panel B. The interaction terms between delayed-reaction factors and dummy of persistent winner/loser: January excluded

	Raw	FF-3	Raw	FF-3	Raw	FF-3	Raw	FF-3	Raw	FF-3	Raw	FF-3
D_{PW}	0.5157*** (5.02)	0.5634*** (5.92)	0.6720*** (3.52)	0.7271*** (4.87)	0.5857*** (5.25)	0.6438*** (6.01)	0.8881*** (4.74)	0.9363*** (6.24)	0.6699*** (3.63)	0.7292*** (4.95)	0.5521*** (2.86)	0.6029*** (4.24)
D_{PL}	-0.9547*** (-6.60)	-1.0361*** (-8.41)	-0.9679*** (-5.29)	-1.1139*** (-7.38)	-1.0957*** (-9.65)	-1.1773*** (-11.24)	-0.8445*** (-4.83)	-0.9883*** (-6.72)	-0.9858*** (-5.42)	-1.1306*** (-7.33)	-0.8856*** (-5.39)	-1.0227*** (-7.37)
$D_{PW} \times D_{Ivol}$	0.2253* (1.84)	0.2591*** (2.91)										
$D_{PL} \times D_{Ivol}$	-0.6305*** (-3.39)	-0.6020*** (-3.31)										
$D_{PW} \times D_{IO}$			-0.2968 (-1.07)	-0.2844 (-1.12)								
$D_{PL} \times D_{IO}$			-1.0395*** (-3.48)	-0.8436*** (-2.81)								
$D_{PW} \times D_{TR}$					-0.0064 (-0.04)	0.0258 (0.18)						
$D_{PL} \times D_{TR}$					-0.3795** (-2.18)	-0.3706** (-2.40)						
$D_{PW} \times D_{SUV}$							0.4093*** (5.96)	0.4199*** (6.48)				
$D_{PL} \times D_{SUV}$							-0.7834*** (-6.10)	-0.7859*** (-6.25)				
$D_{PW} \times D_{DTO}$									0.1360 (1.07)	0.0774 (0.65)		
$D_{PL} \times D_{DTO}$									-0.2073** (-2.04)	-0.1861* (-1.90)		
$D_{PW} \times D_{Dispersion}$											0.2046 (1.06)	0.0485 (0.27)
$D_{PL} \times D_{Dispersion}$											-0.8388*** (-3.82)	-0.9946*** (-5.03)

Table 9. Characteristics of persistent winner and loser stocks

This table presents the time-series averages of estimates from probit regression models. The cross-sectional regression is estimated for each month during the period from 1980 to 2013, with all of winner and loser stocks listed on NYSE, AMEX, or NASDAQ. Coefficients estimated from the cross-sectional regressions are then averaged over the months. The dependent variable is 1 if a winner (loser) stock has a nonzero duration, and 0 otherwise. The duration for a winner (loser) stock is defined as the number of months that a winner (loser) stock is continuously ranked in the top (bottom) decile portfolio in the months after the formation date. The independent variables are firm size, book-to-market ratio (*BM*), idiosyncratic volatility (*Ivol*), institutional ownership (*IO*), monthly turnover (*Turnover*), standardized unexplained volume (*SUV*), change in turnover (*DTO*), previous six-month return ($R_{t-6,t}$), one-month post-formation return ($R_{t,t+1}$), and dispersion in analysts' forecasts (*Dispersion*). Firm size is the logarithm of a stock's market capitalization. *BM* is defined as a firm's book common equity for the fiscal year ending in year $t - 1$ divided by its market equity at the end of December in year $t - 1$. *Ivol* is the residual variance from regressing a firm's daily excess returns on market daily excess returns over the previous 12 months. *IO* is the percentage of outstanding shares held by institutional investors. *Turnover* is the monthly stock turnover ratio of the monthly trading volume to the total number of shares outstanding. *SUV* is a standardized prediction error calculated by a regression of trading volume on the absolute value of stock returns. *DTO* is the difference between a firm's average daily market-adjusted turnover for the month and the average daily market-adjusted turnover over the previous year. *Dispersion* is defined as the standard deviation in analysts' forecasts for stocks in each month. Average monthly firm numbers, *N*, and its percentage are also reported in the bottom of the table. The *t*-statistics calculated from the time series are shown in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively.

	Winners							Losers						
<i>Intercept</i>	-0.3500*** (-8.75)	-0.4676*** (-9.10)	-1.9049*** (-3.57)	-0.5649*** (-11.17)	-4.4762*** (-4.58)	-0.9736*** (-17.67)	-0.4642*** (-10.56)	-1.1801*** (-12.95)	-1.5509*** (-12.78)	-1.2178*** (-12.72)	-2.5133*** (-3.43)	-11.233** (-2.07)	-3.3020*** (-3.93)	-1.7742*** (-7.15)
<i>ln(Size)</i>	-0.0193*** (-2.64)	-0.0183*** (-2.61)	0.0412 (0.48)	-0.0238*** (-3.21)	-0.0705*** (-4.34)	-0.0217*** (-3.57)	-0.0364*** (-6.43)	0.0243* (1.92)	0.0391*** (2.78)	0.0297** (2.15)	0.1455 (1.48)	-0.5847 (-1.48)	0.1427 (1.41)	0.0613* (1.79)
<i>BM</i>	-0.0126 (-1.18)	-0.0036 (-0.31)	0.1792* (1.83)	0.0132 (1.28)	0.1531 (1.50)	0.0314*** (2.84)	0.0054 (0.60)	-0.0952*** (-7.92)	-0.0852*** (-6.66)	-0.0987*** (-7.18)	-0.1091*** (-4.03)	-0.0965 (-0.09)	-0.0679 (-0.80)	-0.1228*** (-4.54)
<i>Ivol</i>		3.6922*** (5.35)		2.8751*** (5.13)	4.2539*** (3.50)	2.8666*** (4.79)	4.2229*** (6.77)		5.5283*** (5.11)		6.8153*** (5.57)	-1.7089 (-0.02)	3.4866** (2.17)	4.9588*** (4.50)
<i>IO</i>			-5.8036* (-1.73)	-0.0773* (-1.73)	-0.8584*** (-2.78)	-0.0311 (-0.98)	-0.0222 (-0.74)			-0.1301* (-1.82)	-0.8020 (-1.57)	-6.5745** (-2.35)	-0.9048** (-2.24)	-0.2656* (-1.90)
<i>Turnover</i>				0.1049*** (7.43)							0.0990* (1.75)			
<i>SUV</i>					0.0842** (2.20)							0.4541* (1.80)		
<i>DTO</i>						0.0829*** (6.15)							0.3397** (2.12)	
<i>Dispersion</i>							13.6261* (1.66)							10.7693*** (2.63)
$R_{t-6,t}$	10.2085*** (18.89)	9.3671*** (21.08)	27.0883*** (3.85)	10.2652*** (26.34)	41.3644*** (8.52)	14.6524*** (30.13)	9.1998*** (31.46)	-12.400*** (-19.11)	-13.350*** (-17.26)	-12.940*** (-18.57)	-19.471*** (-5.14)	-120.91*** (-3.78)	-30.559*** (-5.76)	-14.506*** (-16.54)
$R_{t,t+1}$	0.1627** (1.97)	0.1123** (2.15)	1.5048** (2.09)	0.1439*** (2.63)	1.8739** (2.25)	0.1704*** (3.15)	0.1370*** (3.09)	-0.3672*** (-5.16)	-0.4412*** (-5.71)	-0.3977*** (-5.14)	-0.5948*** (-5.37)	-11.471*** (-2.97)	-0.9083** (-1.99)	-0.4845*** (-5.36)
Log-Likelihood	-158.05*** (-38.20)	-159.32*** (-40.59)	-44.254*** (-18.48)	-134.11*** (-52.38)	-34.639*** (-30.03)	-127.74*** (-55.04)	-157.65*** (-70.95)	-66.096*** (-20.34)	-63.809*** (-19.39)	-64.803*** (-20.14)	-58.975*** (-33.77)	-12.133*** (-17.02)	-49.935*** (-30.50)	-62.059*** (-35.53)
Num. months	384	384	384	384	384	384	384	384	384	384	384	384	384	384
N>0 (%)	323.32 [60.71%]	323.32 [60.71%]	323.32 [60.71%]	323.32 [60.71%]	323.32 [60.71%]	323.32 [60.71%]	96.99 [61.19%]	227.14 [58.80%]	227.14 [58.80%]	227.14 [58.80%]	227.14 [58.80%]	227.14 [58.81%]	227.14 [58.82%]	39.68 [55.47%]
N=0 (%)	209.24 [39.29%]	209.24 [39.29%]	209.24 [39.29%]	209.24 [39.29%]	209.24 [39.29%]	209.24 [39.29%]	61.51 [38.81%]	158.8 [41.20%]	158.8 [41.20%]	158.8 [41.20%]	158.8 [41.20%]	159.8 [41.21%]	160.8 [41.22%]	31.86 [44.53%]

Table 10. Determinants for momentum persistency

This table presents the time-series averages of estimates from the duration model. The cross-sectional duration models are estimated for each month during the period from 1980 to 2013, with all of nonoverlapped persistent winners and losers listed on NYSE, AMEX, or NASDAQ. Coefficients estimated from the cross-sectional duration models are then averaged over the months. The models are as follows:

$$\text{Duration}_{i,t} = a_t + b_{1,t} \ln(\text{Size})_{i,t} + b_{2,t} \text{BM}_{i,t} + b_{3,t} \text{Ivol}_{i,t} + b_{4,t} \text{IO}_{i,t} + b_{5,t} \text{Turnover}_{i,t} + b_{6,t} \text{SUV}_{i,t} + b_{7,t} \text{DTO}_{i,t} \\ + b_{8,t} \text{Dispersion}_{i,t} + b_{9,t} |R_{i,t-6,t}| + b_{10,t} |R_{i,t-6,t}| \times D_{w,t} + b_{11,t} |R_{i,t,t+1}| + b_{12,t} |R_{i,t,t+1}| \times D_{w,t} + \varepsilon_{i,t}$$

The dependent variable is the duration of persistent winners or losers. The duration for a winner (loser) stock is defined as the number of months that a winner (loser) stock is continuously ranked in the top (bottom) decile portfolio in the months after the formation date. The independent variables are firm size, book-to-market ratio (*BM*), idiosyncratic volatility (*Ivol*), institutional ownership (*IO*), monthly turnover (*Turnover*), standardized unexplained volume (*SUV*), change in turnover (*DTO*), dispersion in analysts' forecasts (*Dispersion*), absolute value of previous six-month return ($|R_{t-6,t}|$), absolute value of one-month post-formation return ($|R_{t,t+1}|$), absolute value of previous six-month return times winner dummy, and absolute value of one-month post-formation return times winner dummy. Firm size is the logarithm of a stock's market capitalization. *BM* is defined as a firm's book common equity for the fiscal year ending in year $t - 1$ divided by its market equity at the end of December in year $t - 1$. *Ivol* is the residual variance from regressing a firm's daily excess returns on market daily excess returns over the previous 12 months. *IO* is the percentage of outstanding shares held by institutional investors. *Turnover* is the monthly stock turnover ratio of the monthly trading volume to the total number of shares outstanding. *SUV* is a standardized prediction error calculated by a regression of trading volume on the absolute value of stock returns. *DTO* is the difference between a firm's average daily market-adjusted turnover for the month and the average daily market-adjusted turnover over the previous year. *Dispersion* is defined as the standard deviation in analysts' forecasts for stocks in each month. Winner dummy, D_w , is equal to 1 if the observation belongs to a persistent winner, and 0 otherwise. Average monthly firm numbers, N , is also reported in the bottom of the table. The t -statistics calculated from the time series are shown in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively.

	Model 1	Model 2	Model 3	Model 4
<i>Intercept</i>	0.5481*** (14.90)	0.0461 (0.48)	0.1852*** (2.97)	0.0959* (1.88)
$\ln(\text{Size})$	-0.0086** (-1.98)	-0.0212*** (-3.07)	0.0014 (0.28)	0.0004 -0.1
<i>BM</i>	0.0016 -0.15	0.0082 (0.30)	-0.0315** (-2.00)	-0.0169 (-1.40)
<i>Ivol</i>	5.2386*** (9.19)	11.4028*** (9.44)	8.1811*** (10.69)	8.9769*** (14.09)
<i>IO</i>	-0.0738** (-2.50)	-0.0738* (-1.69)	-0.0597* (-1.72)	0.0161 (0.68)
<i>Turnover</i>	0.0911*** (9.25)			
<i>SUV</i>		0.0281*** (3.82)		
<i>DTO</i>			0.2293*** (11.29)	
<i>Dispersion</i>				1.5564*** (3.98)
$ R_{t-6,t} $	4.2006*** (11.93)	13.4770*** (7.50)	12.2951*** (9.98)	11.9090*** (12.98)
$ R_{t-6,t} \times D_w$	-0.8833*** (-2.69)	-4.8641*** (-3.37)	-5.3004*** (-5.42)	-2.4807*** (-3.13)
$ R_{t,t+1} $	1.1394*** (9.13)	0.6964*** (3.62)	1.5129*** (11.70)	1.1794*** (12.39)
$ R_{t,t+1} \times D_w$	0.3401** (2.00)	0.6335*** (2.85)	0.0718 (0.38)	0.2223** (2.12)
<i>N</i>	123.05	123.05	123.05	102.38
<i>Log-Likelihood</i>	-119.65*** (-66.09)	-72.644*** (-29.33)	-78.564*** (-57.13)	-125.83*** (-85.57)

Table 11. Delayed price adjustment for persistent winner and loser portfolio

This table presents the time-series averages of estimates from the duration model. The cross-sectional duration models are estimated for each month during the period from 1980 to 2013, with all of nonoverlapped persistent winners and losers listed on NYSE, AMEX, or NASDAQ. Coefficients estimated from the cross-sectional duration models are then averaged over the months. The models are as follows:

$$\begin{aligned} \text{Duration}_{i,t} = & a_t + b_{1,t} \ln(\text{Size})_{i,t} + b_{2,t} \text{BM}_{i,t} + b_{3,t} \text{Ivol}_{i,t} + b_{4,t} \text{IO}_{i,t} + b_{5,t} \text{Turnover}_{i,t} + b_{6,t} |R_{i,t,t+1}| + b_{7,t} |R_{i,t-6,t}| \\ & + b_{8,t} |R_{i,t-6,t}| \times \text{Turnover}_{i,t} + b_{9,t} |R_{i,t-6,t}| \times \text{Ivol}_{i,t} + b_{10,t} |R_{i,t-6,t}| \times \text{IO}_{i,t} + \varepsilon_{i,t} \end{aligned}$$

The dependent variable is the duration of persistent winners or losers. The duration for a winner (loser) stock is defined as the number of months that a winner (loser) stock is continuously ranked in the top (bottom) decile portfolio in the months after the formation date. The independent variables are firm size, book-to-market ratio (*BM*), idiosyncratic volatility (*Ivol*), institutional ownership (*IO*), monthly turnover (*Turnover*), absolute value of one-month post-formation return ($|R_{i,t,t+1}|$), absolute value of previous six-month return ($|R_{i,t-6,t}|$), absolute value of one-month post-formation return times monthly turnover, absolute value of one-month post-formation return times idiosyncratic volatility, and absolute value of one-month post-formation return time institutional ownership. Firm size is the logarithm of a stock's market capitalization. *BM* is defined as a firm's book common equity for the fiscal year ending in year $t - 1$ divided by its market equity at the end of December in year $t - 1$. *Ivol* is the residual variance from regressing a firm's daily excess returns on market daily excess returns over the previous 12 months. *IO* is the percentage of outstanding shares held by institutional investors. *Turnover* is the monthly stock turnover ratio of the monthly trading volume to the total number of shares outstanding. *SUV* is a standardized prediction error calculated by a regression of trading volume on the absolute value of stock returns. Average monthly firm numbers, *N*, is also reported in the bottom of the table. The *t*-statistics calculated from the time series are shown in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively.

	Model 1	Model 2	Model 3
<i>Intercept</i>	0.4794*** (9.84)	0.4424*** (5.76)	0.5714*** (10.85)
<i>ln(Size)</i>	-0.0096** (-2.18)	-0.0080* (-1.82)	-0.0066 (-1.46)
<i>BM</i>	0.0008 (0.07)	-0.0002 (-0.01)	-0.0034 (-0.29)
<i>Ivol</i>	7.1825*** (11.46)	10.6688*** (5.25)	7.3062*** (11.81)
<i>IO</i>	-0.0984*** (-3.40)	-0.1186*** (-4.64)	-0.3021*** (-2.88)
<i>Turnover</i>	0.1997*** (4.95)	0.1150*** (10.56)	0.1100*** (10.12)
$ R_{i,t,t+1} $	1.1050*** (21.01)	1.1093*** (20.78)	1.0799*** (20.10)
$ R_{i,t-6,t} $	4.4283*** (7.89)	4.6394*** (4.32)	3.9352*** (7.58)
$ R_{i,t-6,t} \times \text{Turnover}$	0.9392** (2.00)		
$ R_{i,t-6,t} \times \text{Ivol}$		38.8014* (1.77)	
$ R_{i,t-6,t} \times \text{IO}$			-3.0856** (-2.06)
<i>N</i>	123.05	123.05	123.05
<i>Log-Likelihood</i>	-120.95*** (-67.15)	-120.87*** (-67.06)	-120.82*** (-66.63)

Table A.1 Turnover rates for winner and loser stocks: 20 groups

This table presents the average turnover rates for winner and loser stocks in the next formation month. At the end of each month, winner stocks are those stocks with top 5% previous J -month ($J = 3, 6, 9$, and 12) returns, while loser stocks are those stocks with bottom 5% previous J -month returns. Panel A presents average percentages of winner stocks that will be ranked and assigned to 1 of 20 groups in the next formation month. Panel B presents average percentages of loser stocks that will be ranked and assigned to 1 of 20 groups in the next formation month.

Panel A: Winner portfolios

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14	P15	P16	P17	P18	P19	P20
J	(Loser)																			(Winner)
3	1.62%	1.42%	1.42%	1.37%	1.28%	1.34%	1.39%	1.41%	1.52%	1.62%	1.81%	2.04%	2.35%	2.84%	3.59%	4.75%	6.67%	10.88%	20.56%	30.12%
6	1.02%	0.72%	0.71%	0.69%	0.68%	0.68%	0.74%	0.82%	0.90%	0.98%	1.04%	1.33%	1.54%	1.98%	2.68%	3.81%	6.01%	11.80%	29.41%	32.45%
9	1.08%	0.66%	0.69%	0.60%	0.61%	0.61%	0.62%	0.71%	0.74%	0.74%	0.98%	1.06%	1.24%	1.59%	2.16%	3.02%	5.28%	11.01%	33.83%	32.76%
12	1.29%	0.72%	0.61%	0.60%	0.62%	0.61%	0.62%	0.70%	0.64%	0.76%	0.84%	0.90%	1.15%	1.40%	1.94%	2.73%	4.73%	10.50%	36.34%	32.32%

Panel B: Loser portfolios

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14	P15	P16	P17	P18	P19	P20
J	(Loser)																			(Winner)
3	37.02%	35.49%	5.94%	3.59%	2.65%	1.90%	1.52%	1.25%	1.10%	1.03%	0.92%	0.85%	0.84%	0.79%	0.77%	0.78%	0.77%	0.82%	0.86%	1.11%
6	49.13%	30.28%	5.47%	2.73%	1.69%	1.25%	0.98%	0.82%	0.73%	0.71%	0.68%	0.59%	0.56%	0.58%	0.57%	0.54%	0.61%	0.60%	0.62%	0.86%
9	54.52%	27.53%	4.50%	2.18%	1.31%	0.99%	0.81%	0.69%	0.66%	0.61%	0.61%	0.58%	0.56%	0.57%	0.55%	0.56%	0.58%	0.60%	0.65%	0.94%
12	57.32%	25.53%	3.88%	1.87%	1.21%	0.91%	0.83%	0.75%	0.67%	0.58%	0.64%	0.60%	0.58%	0.57%	0.61%	0.61%	0.56%	0.59%	0.67%	1.02%

Table A.2 Information diffusion for persistent winner and loser stocks (durations greater than six months)

This table presents the estimates from the duration model with year and industry fixed effects. The regressions are estimated for all nonoverlapped persistent winners and losers with durations greater than six months. The models are as follows:

$$\text{Duration}_{i,t} = a_t + b_{1,t} |R_{i,t,t+1}| + b_{2,t} |R_{i,t-6,t}| + b_{3,t} |R_{i,t-6,t}| \times \text{Turnover}_{i,t} + b_{4,t} |R_{i,t-6,t}| \times \text{Ivol}_{i,t} + b_{5,t} |R_{i,t-6,t}| \times \text{IO}_{i,t} \\ + \text{Year dummies} + \text{Industry dummies} + \text{controls} + \varepsilon_{i,t}.$$

The dependent variable is the duration of persistent winners or losers. The duration for a winner (loser) stock is defined as the number of months that a winner (loser) stock is continuously ranked in the top (bottom) decile portfolio in the months after the formation date. The duration for a winner (loser) stock is defined as the number of months that a winner (loser) stock is continuously ranked in the top (bottom) decile portfolio in the months after the formation date. The independent variables are absolute value of one-month post-formation return ($|R_{i,t,t+1}|$), absolute value of previous six-month return ($|R_{i,t-6,t}|$), absolute value of one-month post-formation return times monthly turnover, absolute value of one-month post-formation return times idiosyncratic volatility, absolute value of one-month post-formation return time institutional ownership, and control variables including firm size, book-to-market ratio (BM), idiosyncratic volatility ($Ivol$), institutional ownership (IO), monthly turnover ($Turnover$). Firm size is the logarithm of a stock's market capitalization. BM is defined as a firm's book common equity for the fiscal year ending in year $t - 1$ divided by its market equity at the end of December in year $t - 1$. $Ivol$ is the residual variance from regressing a firm's daily excess returns on market daily excess returns over the previous 12 months. IO is the percentage of outstanding shares held by institutional investors. $Turnover$ is the monthly stock turnover ratio of the monthly trading volume to the total number of shares outstanding. The t -statistics are shown in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively.

	Model 1	Model 2	Model 3
$ R_{i,t,t+1} $	0.6751*** (9.02)	1.1532*** (15.26)	1.1587*** (14.36)
$ R_{i,t-6,t} $	7.0889*** (19.27)	8.6749*** (26.86)	11.2740*** (21.02)
$ R_{i,t-6,t} \times \text{Turnover}$	0.04946 (0.33)		
$ R_{i,t-6,t} \times \text{Ivol}$		0.3284* (1.76)	
$ R_{i,t-6,t} \times \text{IO}$			-5.1665*** (-4.56)
Controls	Yes	Yes	Yes
Industry dummy	Yes	Yes	Yes
Year dummy	Yes	Yes	Yes
Log-likelihood	-124706	-122004	-99650